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Rabbit Spleen and Medicinal Herbs: Animal Infectious Diseases, Grassroots Communes, and the State in Maoist China

JONGSIK CHRISTIAN YI

SUMMARY: This article explores how Mao-era China responded to major epizootic and zoonotic diseases. It foregrounds a series of patterns in fighting contagious animal diseases—lockdowns, quarantines, disinfection, mass animal vaccination, mass education, and prioritizing the treatment of infected animals over mass culling—which were together called the Comprehensive Prevention and Treatment (CPT). Shedding light on this understudied topic in the fields of the history of medicine and of the People’s Republic of China (PRC), the author argues that it was not the central or provincial governments but rather local communes that led the effort to protect livestock from animal infectious diseases. This article critically demonstrates how the story of the CPT highlights the resilience of communal actors as well as the possibilities and limitations of the Maoist ideal of self-reliance.

KEYWORDS: animal diseases, Maoist China, communal self-reliance, animal vaccinations, animal culling, Comprehensive Prevention and Treatment

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兔脾与草药：毛泽东时代中国的动物传染病、基层公社与国家

JONGSIK CHRISTIAN YI

摘要：本文探讨了毛泽东时代中国如何应对主要的动物流行病和人畜共患病。文章重点介绍了一系列防治动物传染病的模式——封锁、隔离、消毒、大规模动物疫苗接种、大众教育以及优先治疗受感染动物而非大规模扑杀——这些模式统称为“综合防治”（CPT）。本文阐明了医学史和中华人民共和国（PRC）领域中这一未被充分研究的课题，作者认为，并非中央或省级政府，而是地方公社主导了保护牲畜免受动物传染病侵害的工作。本文批判性地展示了综合防治的故事如何突显了公社行动者的韧性以及毛泽东时代自力更生理想的可能性和局限性。

关键词：动物疾病、毛泽东时代中国、公社自力更生、动物疫苗接种、动物扑杀、综合防治

Late at night on September 3, 1973, when the Great Proletarian Cultural Revolution was still ongoing, six healthy plowing cattle were suddenly found dead in Sijiqing Commune near Beijing. The next day, local Communist cadres carefully declined some peasants’ requests to distribute the meat from the carcasses and, instead, called in a commune veterinarian to inspect them. On September 5, the veterinarian dissected the cattle and provisionally diagnosed anthrax as the cause of death. The commune leaders immediately reported it to the district and municipal governments along with biopsy samples. At the same time, they mobilized all available veterinary workers and ordinary residents in the commune to disinfect cowsheds, roads, and wells, carry out quarantines, and vaccinate nearby bovines. Two days later, Beijing’s veterinary authorities confirmed the presence of *Bacillus anthracis*, the bacterium causing anthrax, in the samples. The outbreak was controlled by September 26. Sijiqing Commune did not experience any more animal deaths. Throughout the process, the central and municipal governments, while keeping an eye on what was occurring on the ground, hardly took direct actions. Communal actors were at the forefront of coping with the deadly animal infectious disease, and they managed to achieve their goal without significant aid or subsidies from the upper-level state apparatus.¹

This article explores how Mao-era China responded to major epizootic and zoonotic diseases such as anthrax, swine fever, foot-and-mouth disease (FMD), brucellosis, and so on. There was a general pattern in fighting them—lockdowns, quarantines, disinfection, mass

¹ “Guanyu Haidian qu Sijiqing gongshe xiangshan dadui fasheng gengniu tanjubing ji chuli qingkuang de baogao” [Report on the outbreak of anthrax in plowing cattle and its treatment in Xiangshan Brigade, Shijiqing Commune, Haidian District] (September 26, 1973), Beijing Municipal Archives (hereafter BMA), 092-003-00122-00040.

1973 年 9 月 3 日深夜，在无产阶级文化大革命仍在进行之时，北京附近的四季青公社突然发现六头健康的耕牛死亡。第二天，当地的共产党员干部小心翼翼地拒绝了一些农民分发尸体肉的要求，而是请来了一位公社兽医进行检查。9 月 5 日，兽医解剖了牛，初步诊断死因是炭疽病。公社领导立即向区、市政府报告，并附上活检样本。同时，他们动员公社所有可用的兽医工作者和普通居民对牛棚、道路和水井进行消毒，实施隔离，并为附近的牛接种疫苗。两天后，北京市兽医部门证实样本中存在炭疽杆菌，即引起炭疽病的细菌。疫情于 9 月 26 日得到控制。四季青公社没有再发生动物死亡。在整个过程中，中央和市政府在关注地面情况的同时，几乎没有采取直接行动。公社行动者站在应对致命动物传染病的最前沿，他们设法在没有上级国家机构大量援助或补贴的情况下实现了目标。¹

本文探讨了毛泽东时代中国如何应对炭疽、猪瘟、口蹄疫、布鲁氏菌病等主要动物流行病和人畜共患病。在防治这些疾病方面，存在一种普遍模式——封锁、隔离、消毒、大规模

¹ “关于海淀区四季青公社香山大队发生耕牛炭疽病及处理情况的报告” [（1973年9月26日），北京市档案馆（以下简称BMA），092-003-00122-00040。

animal vaccination, mass education, and prioritizing the treatment of infected animals over mass culling. These techniques were together called the “Comprehensive Prevention and Treatment” (*zonghexing fangzhi*, hereafter CPT). Shedding light on this understudied topic,² I argue that it was not the state but rather local “self-reliant” communes that led the effort to protect livestock from contagious animal diseases.

Historians of modern China have emphasized that self-reliance (*zili gengsheng*) was one of the central features of Maoism. First of all, it was revolutionary China’s geopolitical principle, which meant that the nation should be autonomous from imperial powers. Maoist China promoted this creed to the Global South at the height of the Cold War.³ But more importantly, it was the Chinese Communist Party’s (CCP) style of governance or domestic policy that foregrounded “the proletarian revolutionary spirit of using local methods, starting from scratch, . . . not fearing failure, and overcoming hardships.”⁴ The Chinese Communist emphasis on local self-reliance was historically derived from the plight that the CCP faced

² On the recent works on the history of “human” medicine and diseases in modern China that are not explicitly related to the issues of animal health, see Ruth Rogaski, *Hygienic Modernity: Meanings of Health and Disease in Treaty-Port China* (Berkeley: University of California Press, 2004); Sean Hsiang-lin Lei, *Neither Donkey nor Horse: Medicine in the Struggle over China’s Modernity* (Chicago: University of Chicago Press, 2014); Miriam Gross, *Farewell to the God of Plague: Chairman Mao’s Campaign to Deworm China* (Berkeley: University of California Press, 2016); Mary Augusta Brazelton, *Mass Vaccination: Citizens’ Bodies and State Power in Modern China* (Ithaca, N.Y.: Cornell University Press, 2019); Xiaoping Fang, *China and the Cholera Pandemic: Restructuring Society under Mao* (Pittsburgh: University of Pittsburgh Press, 2021).

³ For example, see Deborah Bräutigam, *Chinese Aid and African Development: Exporting Green Revolution* (London: Macmillan, 1998); Matthew Galway, *The Emergence of Global Maoism: China’s Red Evangelism and the Cambodian Communist Movement, 1949–1979* (Ithaca, N.Y.: Cornell University Press, 2022).

⁴ Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016), 18.

动物疫苗接种、大众教育以及优先治疗受感染动物而非大规模扑杀。这些技术统称为“综合性防治”（*zonghexing fangzhi*，以下简称 CPT）。为阐明这一未被充分研究的课题，²我认为，在保护牲畜免受传染性动物疾病侵害的努力中，主导者并非国家，而是地方“自力更生”的公社。

现代中国历史学家强调，自力更生（*zili gengsheng*）是毛泽东思想的核心特征之一。首先，它是革命中国的地缘政治原则，意味着国家应独立于帝国主义列强。在冷战高峰期，毛泽东时代的中国向全球南方推广了这一信条。³但更重要的是，它是中国共产党（CCP）的治理方式或国内政策，强调“无产阶级革命精神，即采用地方方法，白手起家，……不惧失败，克服困难”。⁴中国共产党对地方自力更生的强调，在历史上源于中国共产党所面临的困境

² On the recent works on the history of “human” medicine and diseases in modern China that are not explicitly related to the issues of animal health, see Ruth Rogaski, *Hygienic Modernity: Meanings of Health and Disease in Treaty-Port China* (Berkeley: University of California Press, 2004); Sean Hsiang-lin Lei, *Neither Donkey nor Horse: Medicine in the Struggle over China’s Modernity* (Chicago: University of Chicago Press, 2014); Miriam Gross, *Farewell to the God of Plague: Chairman Mao’s Campaign to Deworm China* (Berkeley: University of California Press, 2016); Mary Augusta Brazelton, *Mass Vaccination: Citizens’ Bodies and State Power in Modern China* (Ithaca, N.Y.: Cornell University Press, 2019); Xiaoping Fang, *China and the Cholera Pandemic: Restructuring Society under Mao* (Pittsburgh: University of Pittsburgh Press, 2021).

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⁴ Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016), 18.

during the Yan'an period (1935–47) before it seized power. In the context of domestic and international isolation in the barren Northwest and from Chiang Kai-shek's Nationalist government, the United States, and even the Soviet Union, there was no other way for the CCP to depend on indigenous resources.⁵

Even after the CCP defeated the Nationalists in 1949, clashed with the United States during the Korean War (1950–53), and consolidated its power, local self-reliance was continuously valued. This time it was not because the Communist state lacked the material capacity to support local communities but because it set priorities to minimize the investment in the vast rural area. According to historian Jacob Eyferth, what can be called “primitive socialist accumulation” was adopted in Mao-era China. The Communist leaders believed that “wealth needs to be channeled from the non-industrial, non-socialist sector to state-owned urban industry, through a mechanism of non-equivalent exchange.”⁶ The CCP, therefore, collectivized rural people and tied them “ever more closely to the land, making it impossible for them to leave their place of registration” so that the countryside could become more easily exploitable to the state.⁷ In other words, from the perspective of the state, rural China was not the place in which it was willing to invest resources, but the place from which it hoped to collectively extract surplus values for industrialization.⁸

⁵ Ibid., esp. chap. 1. On the Yan'an period, see Mark Selden, *China in Revolution: The Yanan Way Revisited* (1995; London: Routledge, 2015).
⁶ Jacob Eyferth, “State Socialism and the Rural Household: How Women's Handloom Weaving (and Pig-Raising, Firewood-Gathering, Food-Scavenging) Subsidized Chinese Accumulation,” *Internat. Rev. Soc. Hist.* 62, no. 2 (August 2022): 232.
⁷ Ibid., 237.
⁸ For more comprehensive discussions on the nature of the collectivization and its relation to the primitive accumulation in China, see Mark Selden, *The Political Economy of Chinese Development*

在夺取政权之前的延安时期（1935-1947年）。在贫瘠的西北地区，面对国内外的孤立，以及来自蒋介石的国民政府、美国乃至苏联的压力，中共别无选择，只能依靠本土资源。⁵

即使在中共于1949年击败国民党、在朝鲜战争（1950-1953年）期间与美国发生冲突并巩固其权力之后，地方自力更生仍然持续受到重视。这一次，并非因为共产主义国家缺乏支持地方社区的物质能力，而是因为它将优先事项设定为最大限度地减少对广大农村地区的投资。根据历史学家雅各布·艾弗斯（Jacob Eyferth）的说法，毛泽东时代的中国采取了所谓的“原始社会主义积累”。共产主义领导人认为，“财富需要通过非等价交换机制，从非工业、非社会主义部门输送到国有城市工业。”⁶ 因此，中共将农村人口集体化，并将他们“更紧密地束缚在土地上，使他们无法离开户籍所在地”，以便农村更容易被国家剥削。⁷ 换句话说，从国家的角度来看，中国农村并非国家愿意投入资源的地方，而是国家希望从中集体提取剩余价值以实现工业化的地方。⁸

⁵ 同上，特别是第1章。关于延安时期，参见Mark Selden, 《中国革命：延安道路再探》（1995年；伦敦：Routledge出版社，2015年）。⁶ Jacob Eyferth, “国家社会主义与农村家庭：女性手工织布（以及养猪、拾柴、觅食）如何补贴中国积累”，《国际社会史评论》第62卷第2期（2022年8月）：232页。⁷ 同上，237页。⁸ 关于中国集体化性质及其与原始积累关系的更全面讨论，参见Mark Selden, 《中国发展的政治经济学》

Unlike human health, animal health in the rural area was of lesser importance to the state’s priorities. The central and local governments were accordingly incentivized to superimpose the ideal of communal self-reliance upon local communities and encouraged them to take care of livestock with minimized subsidies. What we come to see at least in the realm of animal health is the figure of the state, which was ideologically interventionist yet materially inactive, and that of grassroots communes, which became capable of fighting animal infectious diseases on their own—represented by the institutionalization of the CPT—despite their ideological subordination to the state.⁹

To support these findings and arguments, I mainly refer to primary documents from the Beijing Municipal Archives, Gansu Provincial Archives, Guangxi Zhuang Autonomous Region Archives, and Guangxi Yulin Municipal Archives. Archival sources about rural communities on the outskirts of the capital give us a typical sense of how they responded to animal diseases when the central government’s presence was highly conspicuous. Cases from Guangxi, the southwestern-most provincial-level administrative unit and farthest from the capital, form a comparative pair with sources from Beijing, allowing us to weigh the dynamics between the central government and local communities. Finally, Gansu province

(Armonk, N.Y.: M. E. Sharpe, 1993); Ho-fung Hung, *The China Boom: Why China Will Not Rule the World* (New York: Columbia University Press, 2015); Tiejun Wen, *Ten Crises: The Political Economy of China’s Development, 1949–2020* (Singapore: Palgrave Macmillan, 2021).

⁹ In this vein, this present study echoes the historical insight that the state-society relationship in Mao-era China was not necessarily zero-sum. See, for instance, James Townsend, *Political Participation in Communist China* (Berkeley: University of California Press, 1967); Julia Strauss, “In Search of PRC History,” *Chin. Quart.* 188 (December 2006): 855–69; Philip Huang, “Rethinking ‘the Third Sphere’: The Dualistic Unity of State and Society in China, Past and Present,” *Mod. Chin.* 45, no. 4 (July 2019): 355–91.

与人类健康不同，农村地区的动物健康对国家而言重要性较低。中央和地方政府因此受到激励，将公社自力更生的理想强加于地方社区，并鼓励它们在补贴最少的情况下照料牲畜。我们至少在动物健康领域看到的是，国家在意识形态上是干预主义的，但在物质上是不活跃的；而基层公社，尽管在意识形态上从属于国家，但却能够依靠自身力量对抗动物传染病——这以 CPT 的制度化为代表。⁹

为了支持这些发现和论点，我主要参考了北京市档案馆、甘肃省档案馆、广西壮族自治区档案馆和广西玉林市档案馆的原始文献。关于首都郊区农村社区的档案资料，让我们对中央政府高度显眼时它们如何应对动物疾病有了典型的认识。来自广西（最西南的省级行政单位，离首都最远）的案例与来自北京的资料形成了对比，使我们能够权衡中央政府与地方社区之间的动态。最后，甘肃省

(纽约州阿蒙克：M. E. Sharpe出版社，1993年)；何妨洪，《中国繁荣：中国为何不会统治世界》（纽约：哥伦比亚大学出版社，2015年）；温铁军，《十大危机：1949-2020年中国发展的政治经济学》（新加坡：Palgrave Macmillan出版社，2021年）。⁹ 本研究与此一脉相承，呼应了毛泽东时代中国国家与社会关系并非必然是零和博弈的历史洞见。例如，参见James Townsend，《共产主义中国的政治参与》（伯克利：加州大学出版社，1967年）；Julia Strauss，“寻找中华人民共和国历史”，《中国季刊》188期（2006年12月）：855-69页；Philip Huang，“重新思考‘第三领域’：中国国家与社会的二元统一，过去与现在”，《现代中国》45卷第4期（2019年7月）：355-91页。

was central in the development of veterinary medicine in modern China as its geography includes both staple farming and pastoral areas. Sources from Gansu province therefore inform us of the situation in places where veterinary expertise and experts were regarded to be more crucial than elsewhere.

Collectives, Animal Diseases, and Veterinary Workers

Before delving into the details about the commune-led CPT, this section briefly introduces the Maoist agrarian collectivization and its relations to animal infectious diseases as background. The CCP started agrarian collectivization in 1953, only four years after coming to power in October 1949. The goal was to maximize the state’s capacity to control and extract an agrarian surplus for the sake of primitive accumulation to support rapid industrialization. The Communists first transformed the village of approximatively thirty families and 145 people into the most basic collective unit, called a “mutual aid team,” “elementary cooperative,” or later “production team.” The households in the village could no longer claim private ownership, and the collective became a unit of production and distribution.¹⁰

The scale rose throughout the decade. On average, seven villages (production teams) formed an “advanced cooperative” or a “production brigade” that consisted of about 220 households and 980 people. After 1958, the largest cooperatives, “people’s communes,”

¹⁰ Franz Schurmann, *Ideology and Organization in Communist China* (Berkeley: University of California Press, 1966), 455; Andrew G. Walder, *China under Mao: A Revolution Derailed* (Cambridge, Mass.: Harvard University Press, 2015), 56.

在中国现代兽医学的发展中，甘肃省一直处于核心地位，因为其地理位置兼具农业和牧业区域。因此，来自甘肃省的资料为我们提供了兽医专业知识和专家被认为比其他地方更为关键的地区的情况。

Collectives, Animal Diseases, and Veterinary Workers

在深入探讨公社主导的 CPT 之前，本节将简要介绍毛泽东时代的农业集体化及其与动物传染病的关系作为背景。中共于 1953 年，即 1949 年 10 月执政四年后，开始推行农业集体化。其目标是最大限度地提高国家控制和提取农业剩余的能力，以实现原始积累，支持快速工业化。共产党首先将大约三十户家庭和 145 人的村庄转变为最基本的集体单位，称为“互助组”、“初级社”或后来的“生产队”。村里的农户不再拥有私人所有权，集体成为生产和分配的单位。¹⁰

整个十年间，规模不断扩大。平均而言，七个村庄（生产队）组成一个“高级合作社”或一个“生产大队”，约有220户家庭和980人。1958年后，最大的合作社“人民公社”

¹⁰ 弗朗茨·舒尔曼， 《共产主义中国的意识形态与组织》（伯克利：加州大学出版社，1966），第455页；安德鲁·G·沃尔德， 《毛泽东领导下的中国：一场脱轨的革命》（马萨诸塞州剑桥：哈佛大学出版社，2015）， 第56页。

appeared and boasted an average population of 15,000. This three-level collective structure incorporating the grassroots rural society belonged to a county and was officially differentiated from the lowest state apparatus, the county government. Communes and lower collectives outlived Mao Zedong’s death in 1976 and were maintained until around 1985.¹¹

Not only human peasants but nonhuman animals went through collectivization. Draft animals (oxen, donkeys, horses, mules, camels, and water buffalos) and meat animals (pigs, cows, sheep, goats, chickens, ducks, geese, and rabbits) were expropriated as cooperatives’ public property. Every collective now established its animal farms and selected some peasants as designated caretakers (*siyangyuan*), who were tasked with taking care of the animals. There were three to seven caretakers in a production team, and a caretaker usually looked after twenty to thirty pigs and five bovines or equines.¹² There were fewer veterinarians than caretakers. Each brigade was to have at least one full-time veterinarian. Every commune had a communal veterinary station staffed with five or more veterinarians who were responsible for the health of around fifteen thousand farm animals.¹³ Production teams usually had no veterinarians at all and had to call in brigade veterinarians when one was necessary. Although it cannot be denied that collectivization, especially at its inception and during the Great Leap Famine (1959–61), negatively affected animal well-being,¹⁴ the local veterinary workforce was able to gradually stabilize the system of communal

出现，平均人口达 15,000 人。这种将基层农村社会纳入其中的三级集体结构属于一个县，并与最低一级的国家机构——县政府——正式区分开来。公社和下级集体在毛泽东 1976 年逝世后仍然存在，并一直维持到 1985 年左右。¹¹

不仅人类农民，非人类动物也经历了集体化。役畜（牛、驴、马、骡、骆驼和水牛）和肉畜（猪、牛、羊、山羊、鸡、鸭、鹅和兔）被征用为合作社的公共财产。每个集体都建立了动物农场，并选派了一些农民作为指定的饲养员（*siyangyuan*），负责照料动物。一个生产队有三到七名饲养员，一名饲养员通常照看二三十头猪和五头牛或马。¹² 兽医比饲养员少。每个大队至少要有一名专职兽医。每个公社都有一个公社兽医站，配备五名或更多兽医，负责约一万五千头农场动物的健康。¹³ 生产队通常根本没有兽医，必要时必须请大队兽医。尽管不可否认，集体化，尤其是在其初期和大跃进饥荒（1959-61）期间，对动物福利产生了负面影响，¹⁴ 但地方兽医队伍能够逐步稳定公社制度。

¹¹ Walder, *China under Mao* (n. 10)

¹² Author’s interview with a former caretaker in Bobai county, Guangxi, June 2021.

¹³ Author’s interview with a former commune veterinarian in Langfang, Hebei, May 2021.

¹⁴ Peter Watson Braden, “Serve the People: Bovine Experiences in China’s Civil War and Revolution, 1935–1961” (Ph.D. diss., University of California, San Diego, 2020), chap. 3.

¹¹ Walder, 《毛泽东时代的中国》（注10）¹² 作者于2021年6月在广西博白县采访了一位前饲养员。¹³ 作者于2021年5月在河北廊坊采访了一位前公社兽医。¹⁴ Peter Watson Braden, “为人民服务：1935-1961年中国内战和革命中的牛类经验”（博士论文，加州大学圣地亚哥分校，2020年），第3章。

management of livestock throughout the 1960s.¹⁵

Epizootic and zoonotic diseases were the biggest threat to the health of collective animals and the communal material life that heavily relied upon the animals. Diseases varied and were differently distributed. If China can be divided into agricultural and pastoral regions, in the agricultural regions where poultry, pigs, oxen, water buffalos, and donkeys were widespread, local veterinary workers mainly dealt with avian flu, anthrax, FMD, swine fever, and swine pneumonia. In the grasslands and deserts, horses, camels, yaks, sheep, and goats were more common, and protecting these animals from anthrax, FMD, brucellosis, ovine parasitic diseases, sheep pox, and glanders was the highest priority. Local governments and communes paid particular attention to anthrax, brucellosis, FMD, and swine fever, among others. The first two could also infect human beings, and the last two were most devastating economically.¹⁶

The central government, too, was aware as early as April 1959, at the height of the agricultural collectivization, that the concentration of livestock could make them more vulnerable to a variety of epizootics as it would “increases the chance of infection of diseases

整个 20 世纪 60 年代牲畜的管理。¹⁵

流行性动物疾病和人畜共患病是对集体动物健康和严重依赖动物的公共物质生活构成最大威胁的疾病。疾病种类繁多，分布各异。如果将中国划分为农牧区，在家禽、猪、牛、水牛和驴广泛分布的农区，当地兽医主要处理禽流感、炭疽、口蹄疫、猪瘟和猪肺炎。在草原和沙漠地区，马、骆驼、牦牛、绵羊和山羊更为常见，保护这些动物免受炭疽、口蹄疫、布鲁氏菌病、绵羊寄生虫病、羊痘和鼻疽的侵害是重中之重。地方政府和公社特别关注炭疽、布鲁氏菌病、口蹄疫和猪瘟等疾病。前两种疾病也可能感染人类，后两种疾病在经济上最具破坏性。¹⁶

早在1959年4月，在农业集体化的高潮时期，中央政府就意识到牲畜的集中会使它们更容易受到各种动物流行病的影响，因为这会“增加疾病感染的机会”。

¹⁵ Jongsik Christian Yi, “More-Than-People’s Communes: Veterinary Workers, Nonhuman Animals, and One Health in Mao-Era China” (Ph.D. diss., Harvard University, 2022), esp. chap. 4.

¹⁶ Guangxi sheng nongyeting, “Guangxi sheng 1954 nian jiachu bayou jihua dagang” [Outline of the 1954 plan for livestock protection and breeding in Guangxi province] (1954), Guangxi Zhuang Autonomous Region Archives (hereafter GARA), X163-001-0394-0083; “1958–1962 nian shouyi fangzhi jianyi shuzi” [Proposition for animal disease prevention and treatment in 1958–1962 with numbers] (1966), Gansu Provincial Archives (hereafter GPA), 222-001-0410-0008.

¹⁵ Jongsik Christian Yi, “More-Than-People’s Communes: Veterinary Workers, Nonhuman Animals, and One Health in Mao-Era China” (博士论文，哈佛大学，2022)，特别是第4章。¹⁶ 广西省农业厅，“广西省1954年家畜保佑计划大纲” [Outline of the 1954 plan for livestock protection and breeding in Guangxi province] (1954)，广西壮族自治区档案馆（以下简称GARA），X163-001-0394-0083；“1958–1962年兽医防治建议数字” [Proposition for animal disease prevention and treatment in 1958–1962 with numbers] (1966)，甘肃省档案馆（以下简称GPA），222-001-0410-0008。

from one animal to another.”¹⁷ Consequently, it was emphasized that each communal and subcommunal collective had to proactively prevent and respond to animal diseases within its jurisdiction in the earliest stages of epidemic. The diversity of the environment and animal species across the country’s vast rural area also made this grassroots collective-based approach appear more sensible.¹⁸

But practical considerations were not the only reason that the state preferred the communal initiative to its direct intervention in protecting nonhuman constituents of the People’s Republic. Whereas sustaining the human health was obviously connected to the CCP’s political legitimacy, the life and death of livestock was not. Therefore, the latter was not very high on the state’s list of resource investments, which were prioritized according to the necessity of rapid socialist primitive accumulation.¹⁹ Against this backdrop, what I call local veterinary workers—rank-and-file veterinary cadres, veterinarians, animal disease prevention workers, and animal caretakers—came to the forefront of responding to contagious animal diseases.

¹⁷ Nongyebu xumu shouyiju, *Renmin gongshe shouyi gongzuo shouce* [Handbook on the work of animal husbandry and veterinary medicine in people’s communes] (Beijing: Nongye chubanshe, 1959), 3.
¹⁸ Ibid., 3, 19.
¹⁹ It is well known that most investment went to industry, especially heavy industry, during the Maoist period. In general, agriculture was less prioritized. It received only 7.1 percent of total state investment during the First Five Year Plan (1953–57) and 11.3 percent during the Great Leap Forward (1958–61). I could not find nationwide statistical data regarding what percentage of the total agricultural investment was allocated to the subsection of animal husbandry and veterinary medicine. But I assume that it fell behind agronomy, irrigation, and agricultural mechanization. Barry Naughton, *The Chinese Economy: Adaptation and Growth* (Cambridge, Mass.: MIT Press, 2017), chaps. 4, 6, and 7; Covell Meyskens, *Mao’s Third Front: The Militarization of Cold War China* (Cambridge: Cambridge University Press, 2020), 51.

从一种动物传播到另一种动物。”¹⁷ 因此，强调每个公社和次级公社集体必须在其管辖范围内，在疫情爆发的最早阶段，积极预防和应对动物疾病。全国广阔农村地区环境和动物物种的多样性也使得这种以基层集体为基础的方法显得更为合理。¹⁸

但实际考量并非国家倾向于公社主动而非直接干预保护中华人民共和国非人类成员的唯一原因。维持人类健康显然与中国共产党的政治合法性相关，而牲畜的生死则不然。因此，牲畜在国家资源投资清单上的优先级不高，这些投资是根据快速社会主义原始积累的必要性来优先排序的。¹⁹ 在此背景下，我称之为地方兽医工作者——基层兽医干部、兽医、动物疫病防治人员和动物饲养员——成为应对动物传染病的主力。

¹⁷ 农业部畜牧兽医局，《人民公社兽医工作手册》[Handbook on the work of animal husbandry and veterinary medicine in people’s communes]（北京：农业出版社，1959），3。¹⁸ 同上，3，19。¹⁹ 众所周知，在毛泽东时代，大部分投资都流向了工业，特别是重工业。总的来说，农业受重视程度较低。在第一个五年计划（1953-1957）期间，农业仅占国家总投资的7.1%，在大跃进（1958-1961）期间占11.3%。我无法找到关于畜牧兽医在农业总投资中所占比例的全国性统计数据。但我认为它落后于农学、灌溉和农业机械化。Barry Naughton，《中国经济：适应与增长》（马萨诸塞州剑桥：麻省理工学院出版社，2017），第4、6、7章；Covell Meyskens，《毛泽东的第三战线：冷战中国的军事化》（剑桥：剑桥大学出版社，2020），51。

Veterinary cadres (*shouyi ganbu* or *xumu shouyi ganbu*) were mid- or low-level CCP members who were in charge of livestock farming and veterinary affairs on the ground. They were not veterinarians themselves, but they could diagnose and treat common, mild ailments and were trained in how to handle vaccines, how to give animals injections, the possible immune reactions, and how to care for inoculated animals. More importantly, veterinary cadres dealt with the “political” tasks related to animal disease control such as making specific action plans for lockdown, quarantine, and disinfection; explaining the rationales of such plan to local peasants; and distributing necessary information and related propaganda materials to every corner of the collective.²⁰

It is worth stressing that these veterinary cadres did not necessarily represent the top-down workings of the state. Scholars who work on socialist China’s grassroots society have questioned the view that local individuals and communities were completely overwhelmed by the authoritarian state. Debates about the nature of local Communist cadres, for example, provide one example of this historiographical move. Joseph Esherick argues that it is difficult to conceptualize rank-and-file Communists in the countryside as “an organizational weapon of obedient apparatchiks commanded by the Party Center.”²¹ Rather, they were strategic

²⁰ “Pingle xian nonglin jishu zhidaozhan guanyu 1953nian xumu shouyi gongzuo zongjie” [Final report of Pingle county agricultural and forestry technical guidance station on the work of animal husbandry and veterinary medicine in 1953] (December 24, 1953), GARA, X163-001-0250-0092; “Dali fangzhi zhu chuanranbing de chubu yijian” [Preliminary opinion for trying the best to prevent and treat contagious swine diseases] (1954), GARA, X163-001-0391-0093.

²¹ Joseph W. Esherick, “Ten Theses on the Chinese Revolution,” *Mod. Chin.* 21, no. 1 (January 1995): 45–76, quotation on 61. Philip C. C. Huang similarly presents the “community operation with state assistance” to explain the nature of the lower-level governance in rural China. See Li-An Zhou, “Understanding China: A Dialogue with Philip Huang,” *Mod. Chin.* 45, no. 4 (July 2019): 398–400.

兽医干部（*shouyi ganbu* 或 *xumu shouyi ganbu*）是中低层中共党员，负责基层畜牧养殖和兽医事务。他们本身并非兽医，但能诊断和治疗常见的轻微疾病，并接受过疫苗处理、动物注射、可能的免疫反应以及接种动物护理方面的培训。更重要的是，兽医干部处理与动物疾病控制相关的“政治”任务，例如制定封锁、隔离和消毒的具体行动计划；向当地农民解释此类计划的理由；以及向集体每个角落分发必要的信息和相关宣传材料。²⁰

值得强调的是，这些兽医干部不一定代表国家的自上而下运作。研究社会主义中国基层社会的学者对地方个人和社区完全被专制国家压倒的观点提出了质疑。例如，关于地方共产党员性质的争论就提供了这种史学转变的一个例子。约瑟夫·埃舍里克（Joseph Esherick）认为，很难将农村的普通共产党员视为“党中央指挥的顺从的机关人员的组织武器”。²¹ 相反，他们是战略性的

²⁰ “平乐县农林技术指导站关于1953年畜牧兽医工作总结” [平乐县农林技术指导站关于1953] 畜牧兽医工作的最终报告（1953年12月24日），GARA, X163-001-0250-0092；“大力防治猪传染病的初步意见” [大力防治猪传染病的初步意见]（1954），GARA, X163-001-0391-0093。²¹ Joseph W. Esherick, “中国革命的十个论点”，*Mod. Chin.* 21, 第1期（1995年1月）：45-76，引文在61页。Philip C. C. Huang也提出了“国家协助下的社区运作”来解释中国农村基层治理的性质。参见Li-An Zhou, “理解中国：与Philip Huang的对话”，*Mod. Chin.* 45, 第4期（2019年7月）：398-400。

“straddlers” between roles as state agents and representatives of local communities’ interests, sometimes complying with and sometimes adroitly derailing and undermining directives handed down by the upper levels of government.²² Grassroots cadres in the veterinary realm were no different in this regard. They were, on the one hand, executors of veterinary policies and, on the other, actors who were keen to preserve their communities’ animal workforce for the sake of themselves and their fellow villagers.

Local cadres were initially reliant on a large number of traditional veterinary practitioners from the era before collectivization. This diverse group of folk animal healers were called “Chinese veterinarians” (*zhongshouyi*) as opposed to “Western veterinarians” (*xishouyi*), which broadly referred to Western-trained veterinarians or practitioners of Western veterinary medicine. While they were less familiar with vaccines and serums than acupuncture and moxibustion, they soon became vaccinators after short training sessions, and local cadres were eager to mobilize them in the anti-epizootic campaigns.²³ After the mid-1950s, when the grassroots collective veterinary system was established, there were no longer private folk veterinarians. Former folk veterinarians were now absorbed into commune- or brigade-level veterinary institutions. Thanks to these grassroots experts, communes became

²² Benjamin L. Read, *Roots of the State: Neighborhood Organization and Social Network in Beijing and Taipei* (Stanford, Calif.: Stanford University Press, 2012); Sang-Soo Park, “Neighborhood Space in 1950s Beijing: Urban Governance in the Early PRC” (Harvard-Yenching Institute Working Paper Series, 2015), <https://www.harvard-yenching.org/research/hyi-working-paper-series-park-sang-soo/>. Also see Shu-min Huang, *The Spiral Road: Change in a Chinese Village through the Eyes of a Communist Party Leader* (Boulder, Colo.: Westview, 1989).

²³ Zhao Qingsen, “Kaizhan jiachu fangyi baohu shengkou jiankang” [Carrying out livestock epidemic prevention and protecting the health of farm animals], *Renmin Ribao* [People’s Daily], August 18, 1949.

他们是国家代理人和地方社区利益代表之间的“跨界者”，有时遵守，有时巧妙地阻挠和破坏上级政府下达的指令。²² 兽医领域的基层干部在这方面也不例外。他们一方面是兽医政策的执行者，另一方面又是为了自己和村民的利益，积极保护社区畜力资源的行动者。

地方干部最初依赖集体化前的大量传统兽医。这些形形色色的民间动物治疗师被称为“中兽医”，以区别于“西兽医”，后者泛指受过西方培训的兽医或西兽医学的从业者。虽然他们对疫苗和血清不如对针灸和艾灸熟悉，但经过短期培训后，他们很快就成了疫苗接种员，地方干部也急于动员他们参与抗疫运动。²³ 20世纪50年代中期，基层集体兽医制度建立后，不再有私人性质的民间兽医。以前的民间兽医现在被吸收到公社或大队一级的兽医机构中。多亏了这些基层专家，公社才得以建立。

²² Benjamin L. Read, *Roots of the State: Neighborhood Organization and Social Network in Beijing and Taipei* (Stanford, Calif.: Stanford University Press, 2012); Sang-Soo Park, “Neighborhood Space in 1950s Beijing: Urban Governance in the Early PRC” (Harvard-Yenching Institute Working Paper Series, 2015), <https://www.harvard-yenching.org/research/hyi-working-paper-series-park-sang-soo/>. Also see Shu-min Huang, *The Spiral Road: Change in a Chinese Village through the Eyes of a Communist Party Leader* (Boulder, Colo.: Westview, 1989).²³ 赵青森，“开展家畜防疫保护牲口健康” [开展家畜防疫保护牲口健康], 《人民日报》 [《人民日报》], 1949年8月18日。

able to reproduce the next generations of veterinarians who generally were called collective veterinarians or “barefoot animal doctors,” particularly during the Cultural Revolution (1966–76).²⁴

Grassroots cadres and veterinarians were supported by the even greater number of epizootic prevention workers (*fangyiyuan*) and caretakers. As collectivization intensified, each production team was required to educate at least one passionate young peasant to produce a prevention worker. In this way, by 1960 there were roughly twenty thousand prevention workers in each province.²⁵ These prevention workers would assist cadres in enforcing animal disease control (Figure 1). Meanwhile, although caretakers were not expected to play a particular role in the midst of epidemic, they were at the forefront of everyday prevention efforts, such as keeping individual animals clean and healthy and conducting regular disinfections of collective farms.²⁶ All these different veterinary workers worked in concert with each other to prevent or minimize losses from infectious animal

²⁴ “Xingan xian zhongshouyi tudi zanxing banfa” [Xingan county temporary measures on the apprenticeship of traditional Chinese veterinary medicine], *Minjian shouyi tongxun* 3 (January 1958): 5–8; “Ruhe zhuangdale minjian shouyi duiwu de” [How we cultivated the groups of folk veterinarians] (1958), GPA, 227-001-0410-0016; Qianxian xumu shouyi gongzuozhan, *Chijiao shouyi shouce* [Barefoot veterinarian handbook] (Qianxian: Qianxian hongse yinshuachang, 1971). For more discussions on the better known “barefoot doctors,” see Xiaoping Fang, *Barefoot Doctors and Western Medicine in China* (Rochester, N.Y.: University of Rochester Press, 2012); Daqing Zhang and Paul Unschuld, “China’s Barefoot Doctor: Past, Present, and Future,” *Lancet* 372, no. 9653 (2008): 1865–67; Miriam Gross, “Between Party, People, and Profession: The Many Faces of the ‘Doctor’ during the Cultural Revolution,” *Med. Hist.* 62, no. 3 (July 2018): 333–59.

²⁵ “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushu” [To guarantee the rapid development of swine production, collective pigsties in various places accelerate vaccination], *Renmin Ribao*, April 10, 1960.

²⁶ “Guanyu jiben xiaomie zhuwen de yijian” [Opinion about basically eliminating swine fever] (1964), GPA, 222-001-0346-0005.

能够培养出下一代兽医，他们通常被称为集体兽医或“赤脚兽医”，尤其是在文化大革命（1966-76）期间。²⁴

基层干部和兽医得到了数量更多的防疫员（*fangyiyuan*）和饲养员的支持。随着集体化深入，每个生产队都被要求培养至少一名有热情的年轻农民成为防疫员。通过这种方式，到 1960 年，每个省大约有两万名防疫员。²⁵ 这些防疫员将协助干部执行动物疫病控制（图 1）。同时，尽管饲养员在疫情期间不被期望发挥特殊作用，但他们处于日常预防工作的前沿，例如保持个体动物清洁健康，并对集体农场进行定期消毒。²⁶ 所有这些不同的兽医工作者协同合作，以预防或最大限度地减少动物传染病造成的损失

²⁴ “Xingan xian zhongshouyi tudi zanxing banfa” [Xingan county temporary measures on the apprenticeship of traditional Chinese veterinary medicine], *Minjian shouyi tongxun* 3 (January 1958): 5–8; “Ruhe zhuangdale minjian shouyi duiwu de” [How we cultivated the groups of folk veterinarians] (1958), GPA, 227-001-0410-0016; Qianxian xumu shouyi gongzuozhan, *Chijiao shouyi shouce* [Barefoot veterinarian handbook] (Qianxian: Qianxian hongse yinshuachang, 1971). For more discussions on the better known “barefoot doctors,” see Xiaoping Fang, *Barefoot Doctors and Western Medicine in China* (Rochester, N.Y.: University of Rochester Press, 2012); Daqing Zhang and Paul Unschuld, “China’s Barefoot Doctor: Past, Present, and Future,” *Lancet* 372, no. 9653 (2008): 1865–67; Miriam Gross, “Between Party, People, and Profession: The Many Faces of the ‘Doctor’ during the Cultural Revolution,” *Med. Hist.* 62, no. 3 (July 2018): 333–59.

²⁵ “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushu” [To guarantee the rapid development of swine production, collective pigsties in various places accelerate vaccination], *Renmin Ribao*, April 10, 1960.

²⁶ “Guanyu jiben xiaomie zhuwen de yijian” [Opinion about basically eliminating swine fever] (1964), GPA, 222-001-0346-0005.

diseases throughout the Maoist era.

With the ambiguous guidelines and the lack of substantive support from the state, local veterinary workers, through trial and error, came up with a series of protocols to respond to epizootics throughout the 1950s. They typically included lockdown, quarantine, disinfection, vaccination, treatment, and mass education. In the late 1950s, the central government retroactively recognized these bottom-up actions and christened them “Comprehensive Prevention and Treatment.” Upper-level Communist leaders especially legitimated the CPT in the name of Mao’s famous call for the “mass line,”²⁷ as its grassroots-centered characteristics facilitated the participation of a wide range of local constituents in public duties for the revolution.²⁸ This nationwide paradigm of the CPT remained in place in the 1960s and 1970s without major changes and with few regional variants.

Lockdown, Quarantine, and Disinfection

The most basic step of the CPT was to impose lockdowns and quarantines in infected areas and to disinfect animals, sheds, roads, and vehicles. Maoist China’s early experiences of putting these measures into effect were geographically concentrated in the pastoral regions of

²⁷ On the “mass line,” see Ilpyong Kim, *The Politics of Chinese Communism: Kiangsi under the Soviets* (Berkeley: University of California Press, 1973); Xiaodong Ding, “The People According to the Chinese Communist Party: Reinterpreting the Party’s Political Theory,” *Mod. Chin.* 46, no. 2 (April 2019): 193–223.
²⁸ “Quanguo zhongxi shouyi yanjiu gongzuohuiyi queding” [Determinations of national Chinese and Western veterinary research working conference], *Renmin Ribao*, February 11, 1959; “Guanyu xiaomie shengchu koutiyi gongzuo zongjie baogao” [Final work report on eliminating livestock foot-and-mouth disease] (April 26, 1965), GPA, 128-002-0718-0002.

整个毛泽东时代的疾病。

由于国家指导方针模糊且缺乏实质性支持，地方兽医工作者通过反复试验，制定了一系列应对20世纪50年代流行病的方案。这些方案通常包括封锁、隔离、消毒、疫苗接种、治疗和大众教育。20世纪50年代末，中央政府追溯性地认可了这些自下而上的行动，并将其命名为“综合防治”。上层共产党领导人特别以毛泽东著名的“群众路线”号召²⁷的名义使综合防治合法化，因为其以基层为中心的特点促进了广大地方民众参与革命的公共事务²⁸。这种全国性的综合防治模式在20世纪60年代和70年代没有重大变化，区域差异也很小。

封锁、隔离和消毒

CPT最基本的步骤是在疫区实行封锁和隔离，并对动物、兽棚、道路和车辆进行消毒。毛泽东时代中国在实施这些措施的早期经验主要集中在牧区。

²⁷ 关于“群众路线”，参见 Ilpyong Kim, *The Politics of Chinese Communism: Kiangsi under the Soviets* (Berkeley: University of California Press, 1973); Xiaodong Ding, “The People According to the Chinese Communist Party: Reinterpreting the Party’s Political Theory,” *Mod. Chin.* 46, no. 2 (April 2019): 193–223。²⁸ “全国中西兽医研究工作会议确定” [全国中西兽医研究工作会议的决定], 《人民日报》，1959年2月11日；“关于消灭牲畜口蹄疫工作总结报告” [消灭牲畜口蹄疫的最终工作报告]（1965年4月26日），GPA, 128-002-0718-0002。

the Northwest where nomadic people had relied on animals for generations. As early as March 1951—before local collectives came into being—the Gansu Provincial Government reported to Beijing that while confronting an outbreak of FMD in the province, they established watchhouses and inspection stations on major thoroughfares to control and disinfect passing. They also shut off large waterways. Infected animals were put into quarantine and looked after separately. If the diseases had already spread within a particular herd, the entire herd was separated from the healthy herds and kept in a designated grazing district.²⁹

Some characteristics of this early experience in the Northwest shaped what would become the CPT. The 1951 report, for example, did not take into consideration the mass culling of infected animals. I will come back to this issue of culling in a later section. Here, the more important point is that the approach to animal diseases in the early Maoist years was based not on sheer bureaucratic capacity but on local people’s participation. According to the report, the provincial government’s preventative efforts needed to be “truly combined with [those of] the masses of people.” Also, “traditional animal healers, exemplary model workers, female peasants, shepherds, and so forth could play a central role in epizootic prevention.” Therefore, local leaders “absolutely tried to have them join” the temporary “small group for livestock protection.” It may not be surprising that a nascent government that lacked human and material resources had no choice but to depend on local nonstate actors.³⁰

²⁹ “Jieshao 51 nian koutiyi fangzhi jingyan” [Introducing the experience of preventing and treating FMD in 1951] (April 8, 1952), GPA, 222-002-0068-0001.

³⁰ Ibid.

西北地区，游牧民族世代都依赖动物生存。早在1951年3月——在地方集体经济出现之前——甘肃省政府就向北京报告，他们在应对该省口蹄疫爆发时，在主要交通要道设立了检查站和检疫站，以控制和消毒过往的动物。他们还关闭了大型水路。受感染的动物被隔离并分开照看。如果疾病已在特定畜群中传播，则整个畜群将与健康的畜群分开，并饲养在指定的放牧区。²⁹

西北地区早期经验的一些特点塑造了后来的 CPT。例如，1951 年的报告没有考虑大规模扑杀受感染的动物。我将在后面的章节中回到扑杀问题。这里更重要的一点是，毛泽东时代早期处理动物疾病的方法不是基于纯粹的官僚能力，而是基于当地人民的参与。根据报告，省政府的预防工作需要“真正与[人民群众的]工作相结合”。此外，“传统兽医、模范工人、女农民、牧羊人等可以在动物疫病预防中发挥核心作用”。因此，地方领导人“绝对努力让他们加入”临时的“牲畜保护小组”。一个缺乏人力和物力资源的新生政府别无选择，只能依靠地方非国家行为者，这可能不足为奇。³⁰

²⁹ “介绍51年口蹄疫防治经验” [介绍 1951]口蹄疫防治经验（1952年4月8日），GPA, 222-002-0068-0001。³⁰ 同上。

Reliance on grassroots participation, however, did not disappear as state power stabilized throughout the first decade of the People’s Republic. Rather, as we have already discussed, after the ambitious collectivization and consequential concentration of farm animals made herds more vulnerable to epizootics, high-ranking leaders decided that letting local communities collectively and flexibly respond to animal epidemics would be more efficient than centralized methods.³¹ Moreover, the state was not willing to commit resources to a rural issue that was not regarded as critical for legitimacy. Reflecting the state’s preference for making local communities a primary actor in the fight against animal diseases, a Gansu provincial report on FMD in 1956 criticized the leadership of Linxia county for overestimating their own administrative capacity and not asking folk veterinarians for help. The county leaders were so complacent and lackadaisical that they thought that they could contain the spread of the disease with limited support from a few teenage boys who were asked to assist in the inspection stations.³²

In terms of the response to epizootics in the mid- and late 1950s, the bottom-up initiative ironically pushed in a top-down way led to growing sophistication in related countermeasures on the ground. Folk veterinarians and peasants made clear the point that the hay and waste from an infected animal’s shed had to be incinerated.³³ They suggested that not only the movement of living ill animals but also that of any kinds of animal products

³¹ Nongyebu xumu shouyiju, *Renmin gongshe shouyi gongzuo shouce* (n. 17).
³² “Guanyu koutiyi fangzhi gongzuo di er jieduan de qingkuang baogao” [Situation report on the second stage of the FMD prevention and treatment work] (October 22, 1956), GPA 190-002-0157-0009.
³³ “Dali fangzhi zhu chuanranbing de chubu yijian” (n. 20).

然而，随着中华人民共和国成立后第一个十年国家权力的稳固，对基层参与的依赖并未消失。相反，正如我们已经讨论过的，在雄心勃勃的集体化和随之而来的农场动物集中化使畜群更容易受到流行病影响之后，高级领导人认为，让地方社区集体灵活地应对动物流行病将比集中式方法更有效率。³¹ 此外，国家不愿为不被视为对其合法性至关重要的农村问题投入资源。甘肃省 1956 年一份关于口蹄疫的报告批评临夏县领导高估了自身的行政能力，没有向民间兽医寻求帮助，这反映了国家倾向于让地方社区成为对抗动物疾病的主要参与者。县领导们自满而散漫，他们认为只需少数被要求协助检查站的少年提供有限支持，就能控制疾病的传播。³²

就20世纪50年代中后期对流行病的反应而言，自下而上的举措以自上而下的方式推动，这反而促使相关应对措施在基层日益完善。民间兽医和农民明确指出，必须焚烧受感染动物棚中的干草和粪便。³³ 他们建议，不仅要控制患病活体动物的移动，还要控制任何动物产品的移动

³¹ 农业部畜牧兽医局，《人民公社兽医工作手册》（注17）。³² “关于口蹄疫防治工作第二阶段的情况报告” [口蹄疫防治工作第二阶段情况报告]（1956年10月22日），GPA 190-002-0157-0009。³³ “大力防治猪传染病的初步意见”（注20）。

from infected areas must be rigidly checked. Also, as each collective became the main unit in the fight against epizootic diseases, they came to pay attention to developing a collaborative prevention system “among counties, townships, and collectives.” Before long, these collective-led developments would be officially recognized by the central government as the CPT program.³⁴

The discussion on when lockdowns and quarantines should be lifted mattered. To be certain, both collectives and county- or upper-level governments agreed that lockdown and quarantine “should not be eased too early.” At least when it came to FMD, there seemed to be a consensus as early as 1956 that the end of an outbreak meant the status that “the majority of infected cattle regain their health, *and* that infection no longer increases.” Gansu provincial leaders more carefully legislated that collectives could lift lockdown with the approval of county governments only after fourteen days of having achieved such status.³⁵ Given that the endpoint of lockdown and quarantine was determined according to the double criteria of the recovery of herds and the disappearance of contagion, this consensus reinforced and was legitimized by the idea that the veterinary actors should stress preserving the animal power of a community as much as eliminating all pathogens within the jurisdiction.³⁶

In the early 1960s, the collective-based CPT was further institutionalized. When FMD spread from the Northeast and Inner Mongolia and started to engulf Beijing in late 1963 and early 1964, the division of labor between upper-level governments and grassroots

对来自疫区的检查必须严格。此外，由于每个集体都成为对抗动物传染病的主要单位，它们开始关注发展“县、乡、集体之间”的协作预防体系。不久之后，这些由集体主导的发展将被中央政府正式认可为CPT计划。³⁴

关于何时解除封锁和隔离的讨论很重要。可以肯定的是，集体和县级或更高级别的政府都同意，封锁和隔离“不应过早放松”。至少在口蹄疫方面，早在1956年似乎就已达成共识，即疫情结束意味着“大多数受感染牛只恢复健康，且感染不再增加”的状态。甘肃省领导更谨慎地规定，集体只有在达到这种状态十四天后，经县政府批准才能解除封锁。³⁵ 鉴于封锁和隔离的终点是根据畜群恢复和传染病消失的双重标准确定的，这一共识通过以下理念得到加强和合法化：兽医人员应强调保护社区的畜力，同时消除管辖范围内的所有病原体。³⁶

20世纪60年代初，集体化的CPT得到进一步制度化。1963年末至1964年初，当口蹄疫从东北和内蒙古蔓延并开始席卷北京时，上级政府与基层之间的分工

³⁴ “Guanyu koutiyi fangzhi gongzuo di er jieduan de qingkuang baogao” (n. 32).

³⁵ Ibid., emphasis added.

³⁶ Ibid.

³⁴ “关于口蹄疫防治工作第二阶段的情况报告”（n. 32）。³⁵ 同上，着重部分为译者所加。³⁶ 同上。

collectives emerged. Although the Beijing Municipal Government and Hebei Provincial Government guided the overall response to the outbreak, their direct actions were largely confined to major railway stations, intercity bus terminals, and municipal distribution channels for food and goods. People’s communes and lower collectives on the outskirts of the capital took the lead in protecting their own communities. A large number of commune veterinarians, prevention workers, caretakers, and even militias stood guard at inspection stations along country roads and disinfected collective animal farms.³⁷

Because this bottom-up initiative in deploying lockdown and quarantine generally worked, Communist authorities became even more convinced that each collective, as a unit, could and should continue to be self-reliant in responding to epizootic diseases. And it also seemed that the state was not obligated to materially support the communes. The communes, too, internalized the idea that they should autonomously take care of all the anti–animal disease measures. Hence, although no one disagreed that harmonious intercommunal collaboration in prevention was desirable, it became very common for a collective to blame another for disseminating pathogens by failing to carry out its responsibilities.

When a suburban community in Beijing grappled with swine fever in 1964, for example, its residents pinpointed “a bicycle rider who came from outside to sell pigs” as the origin of the contagion. These and similar grassroots disputes led the central government to

³⁷ “Duiyu Beijing shi fangzhi koutiyi de jinji jianyi” [Urgent proposition of controlling FMD in the city of Beijing] (January 24, 1964), BMA, 010-001-00446-00099; “Guanyu jiaqiang jianyi yanfang jiachu koutiyi chuanru de jinji tongzhi” [Urgent notification on strengthening inspection and quarantine and strictly preventing livestock FMD from spreading] (December 19, 1963), BMA, 119-001-00795-00106.

集体出现了。尽管北京市政府和河北省政府指导了疫情的总体应对，但它们的直接行动主要限于主要火车站、城际巴士总站以及食品和商品的市政分销渠道。首都郊区的人民公社和下级集体率先保护自己的社区。大量的公社兽医、防疫人员、看护人员，甚至民兵，在乡村道路沿线的检查站站岗，并对集体动物农场进行消毒。³⁷

由于这种自下而上部署封锁和隔离的举措普遍奏效，共产主义当局更加确信，每个集体作为一个单位，能够而且应该继续在应对动物流行病方面自力更生。而且，国家似乎也没有义务为公社提供物质支持。公社也内化了它们应该自主处理所有动物疫病防治措施的想法。因此，尽管没有人不同意公社间和谐的协作预防是可取的，但一个集体因另一个集体未能履行职责而传播病原体的情况变得非常普遍。

例如，1964年，当北京郊区的一个社区与猪瘟作斗争时，居民们将“一个从外面来卖猪的骑自行车的人”确定为传染源。这些以及类似的基层纠纷促使中央政府

³⁷ “对于北京市防治口蹄疫的紧急建议” [Urgent proposition of controlling FMD in the city of Beijing] (1964年1月24日)，BMA, 010-001-00446-00099；“关于加强检疫严防家畜口蹄疫传入的紧急通知” [Urgent notification on strengthening inspection and quarantine and strictly preventing livestock FMD from spreading] (1963年12月19日)，BMA, 119-001-00795-00106。

discourage intercollective animal trade and to come up with the slogan “self-breeding and self-raising” (*zifan ziyang*).³⁸ Now provincial and county-level governments expected communes and lower collectives to train some peasants as breeders so that grassroots communities could reproduce and maintain farm animals on their own, without trading with the outside. “Self-breeding and self-raising” also meant that collectives needed to be self-reliant not only in maximizing the number of collective livestock but also in minimizing the loss of animals when epizootic diseases struck. And this demand for communal self-reliance clearly became the core part of the CPT, especially during the last decade of the Maoist era.³⁹

We may assume that the commune-led CPT testifies to the nonstate actors’ subjective capacity under the infamous authoritarian state. Nonetheless, the grassroots veterinary workers themselves were in fact challenged by the ideal of self-reliance. The goal was always difficult to achieve. After major epizootics, it turned out that collectives needed help from the outside to recoup the loss of livestock. Some pig-raising collectives in Guangxi “relied on the [outside] market to buy new pigs” to refill their collective farms hit by another severe swine fever outbreak in 1972. Unfortunately, the newly acquired pigs brought about a new spike of infections. What the upper-level governments, which were ideologically rich yet materially inactive, did was to simply criticize particular collectives for violating the principle of self-

³⁸ “Beijing shi xumu shouyi xuehui juxing zhuwen fangzhi zuotanhui jianbao” [Summary report on the swine fever prevention and treatment symposium organized by Beijing association for animal husbandry and veterinary medicine research] (July 28, 1964), BMA, 010-001-00187-00055.

³⁹ “Zhonggong zhongyang, guowuyuan guanyu fazhan dashengchu de jixiang guiding” [The Central Committee of the CCP and State Council’s several regulations for the development of big livestock breeding] (November 22, 1962), in *Xumu shouyi gongzuo cankao ziliao*, ed. Gansu sheng xumuting (Lanzhou: Gansu sheng xumuting, 1964), 1–3.

劝阻集体间的牲畜交易，并提出“自繁自养”的口号 (*zifan ziyang*)。³⁸ 现在，省县两级政府要求公社和下级集体培训一些农民成为饲养员，以便基层社区能够自行繁殖和饲养牲畜，而无需与外界进行交易。“自繁自养”也意味着集体不仅要在最大限度地增加集体牲畜数量方面自给自足，而且要在疫病爆发时最大限度地减少牲畜损失方面自给自足。这种对公社自力更生的要求显然成为CPT的核心部分，尤其是在毛泽东时代的最后十年。³⁹

我们或许会认为，公社主导的CPT证明了在臭名昭著的威权国家下非国家行为者的主观能力。然而，基层兽医工作者本身却受到了自力更生理想的挑战。这个目标总是难以实现。在重大流行病爆发后，事实证明，集体需要外部帮助才能弥补牲畜损失。广西的一些养猪集体“依靠[外部]市场购买新猪”，以补充其在1972年遭受另一次严重猪瘟疫情袭击的集体农场。不幸的是，新购买的猪带来了新的感染高峰。上级政府在思想上富足但在物质上不活跃，他们所做的只是简单地批评某些集体违反了自力更生的原则。

³⁸ “北京市畜牧兽医学会举行猪瘟防治座谈会简报” [（1964年7月28日），北京市档案馆，010-001-00187-00055]。³⁹ “中共中央、国务院关于发展大牲畜的几项规定” [（1962年11月22日），载甘肃省畜牧厅编：《畜牧兽医工作参考资料》（兰州：甘肃省畜牧厅，1964年），第1-3页。

breeding and self-raising.⁴⁰ Under the paradigm of the CPT, local veterinary cadres and workers were forced to bear the heavy burden of protecting nonhuman collective members by hook or by crook.

Mass Animal Vaccination

Lockdown, quarantine, and disinfection were the first actions that local veterinary workers took when animal infectious diseases broke out. But however meticulously these measures were carried out, there were always casualties, for the most part livestock but also sometimes human. For instance, when anthrax swept Jianling Brigade, Guangxi, in 1962, fifteen oxen and eighteen people contracted the disease; ten oxen and three people died. Three pigs and a dog also died from licking up the infected bovines’ guts and bone marrow.⁴¹ For a community of less than a thousand people, this was not insignificant damage. As such, prevention was highly emphasized as part of the CPT program, which came to produce another famous slogan: “prioritizing prevention over treatment” (*fang zhongyu zhi*). And for prevention, there was no better option than mass vaccination.⁴²

Mass animal vaccination required mass education. It was different than bringing local villagers to inspection stations and asking them to control and disinfect the traffic of

这是预印本，已接受的文章计划刊登于《医学史通报》第98卷第2期（2024年夏季）。本文已进行编辑，但尚未分页。可能还会进行进一步编辑。请留意最终文章发布详情。

繁殖和自养。⁴⁰ 在CPT的范式下，地方兽医干部和工作人员被迫不择手段地承担起保护非人类集体成员的重担。

大规模动物疫苗接种

封锁、隔离和消毒是地方兽医工作者在动物传染病爆发时采取的首要行动。但无论这些措施执行得多么细致，总会有伤亡，大部分是牲畜，有时也包括人类。例如，1962年炭疽病席卷广西建陵大队时，有15头牛和18人感染了疾病；10头牛和3人死亡。三头猪和一条狗也因舔食受感染牛的内脏和骨髓而死亡。⁴¹ 对于一个不足千人的社区来说，这不是微不足道的损失。因此，预防作为CPT项目的一部分受到了高度重视，并由此产生了另一个著名的口号：“预防重于治疗”（*fang zhongyu zhi*）。而对于预防，没有比大规模疫苗接种更好的选择。⁴²

大规模动物疫苗接种需要大规模的宣传教育。这与将当地村民带到检查站，要求他们控制和消毒交通不同。

⁴⁰ “Guanyu shengzhu yibing wenti de baogao” [Report on the problems of contagious swine diseases] (October 16, 1972), Guangxi Yulin Municipal Archives (hereafter GYMA), 064-001-012-09.

⁴¹ “Guanyu Pingnan xian Wujiang deng xiang fasheng niu tanjubing ganran renchu bingwang de tongbao” [Notification on the outbreak of bovine anthrax and the death of infected people and animals in Wujiang and other townships in Pingnan county] (December 15, 1962), GYMA, 030-009-013-04.

⁴² Cai Ziwei, “Jiji cujin xumuye shengchan dayuejin” [Actively promoting the great leap forward in animal production], *Renmin Ribao*, June 4, 1958.

⁴⁰ “关于生猪疫病问题的报告” [Report on the problems of contagious swine diseases] (1972年10月16日)，广西玉林市档案馆（以下简称GYMA），064-001-012-09。⁴¹ “关于平南县武江等乡发生牛炭疽病感染人畜病亡的通报” [Notification on the outbreak of bovine anthrax and the death of infected people and animals in Wujiang and other townships in Pingnan county] (1962年12月15日)，GYMA, 030-009-013-04。⁴² 蔡子伟，“积极促进畜牧业生产大跃进” [Actively promoting the great leap forward in animal production], 《人民日报》，1958年6月4日。

people and animals over a certain period of time. If local collectives were to administer vaccines, someone present needed to have some practical knowledge, such as how to discern the epizootic disease in question from other disorders, how to store vaccines appropriately, how to determine the correct dosage, how to restrain animals while giving a shot, how to be aware of any possible immune reactions, and so on. Ideally, someone in the collectives should know who Louis Pasteur was and how vaccinations actually work. The story of the CPT, as it included mass animal vaccination, was that of the broad circulation of knowledge across different social stratum.⁴³

Local veterinary cadres played an integral role in this process of “gradually fostering the scientific prevention practice of the masses” (*zhujian yangcheng qunzhong kexuede fangyi xiguan*). In fact, they themselves were beneficiaries of mass education. In 1951–53, the first cohorts of local Communist veterinary cadres were produced through short-term training organized by provincial or county-level governments. They were not regarded as experts in healing animals, but they were intensively introduced to the “common senses regarding scientific disease prevention and therapeutics that the masses could easily get” and the handling of “preventative materials.”⁴⁴

In the 1950s and after, while utilizing their knowledge and leading epizootic

⁴³ In this regard, the story of animal mass vaccination can be related to the recent historiographical focus on the Maoist ideal of mass science, the mass-oriented policy of knowledge production and its dissemination. On the self-reliant and antitechnocratic nature of mass science, see Schmalzer, *Red Revolution, Green Revolution* (n. 4). As for mass science’s central principle of “experiment without contingency,” which means that working people were bound to succeed in producing and accumulating “correct” scientific knowledge, see Rui Kunze and Marc Andre Matten, *Knowledge Production in Mao-Era China: Learning from the Masses* (Lanham, Md.: Lexington Books, 2021).

⁴⁴ “Dali fangzhi zhu chuanranbing de chubu yijian” (n. 20).

在特定时期内，人与动物之间存在着某种联系。如果地方集体要接种疫苗，在场的人就需要具备一些实践知识，例如如何区分所讨论的动物流行病与其他疾病，如何妥善储存疫苗，如何确定正确的剂量，如何在注射时控制动物，如何注意任何可能的免疫反应等等。理想情况下，集体中的某个人应该知道路易·巴斯德是谁以及疫苗接种的实际工作原理。CPT 的故事，包括大规模动物疫苗接种，是知识在不同社会阶层广泛传播的故事。⁴³

地方兽医干部在“逐渐养成群众科学的防疫习惯”（*zhujian yangcheng qunzhong kexuedefangyi xiguan*）这一过程中发挥了不可或缺的作用。事实上，他们自己就是群众教育的受益者。1951-53 年，第一批地方共产主义兽医干部通过省或县级政府组织的短期培训培养出来。他们不被视为动物治疗专家，但他们被集中介绍了“群众容易掌握的科学防疫和治疗常识”以及“预防材料”的处理。⁴⁴

In the 1950s and after, while utilizing their knowledge and leading epizootic

⁴³ 在这方面，动物大规模疫苗接种的故事可以与最近史学界对毛泽东时代大众科学理想、以大众为导向的知识生产及其传播政策的关注联系起来。关于大众科学的自力更生和反技术官僚性质，请参见 Schmalzer, 《红色革命，绿色革命》（注 4）。至于大众科学“无偶发性实验”的核心原则，即劳动人民注定会成功生产和积累“正确”的科学知识，请参见 Rui Kunze 和 Marc Andre Matten, 《毛泽东时代中国的知识生产：向群众学习》（马里兰州兰哈姆：列克星敦图书，2021 年）。⁴⁴ “大力防治猪传染病的初步意见”（注 20）。

prevention within their collectives, veterinary cadres were keen on recruiting “activists” (*jiji fenzi*), passionate villagers who showed strong support for and loyalty to the Communist Party. They encouraged these activists to participate in the regular vaccinating of livestock and gave them a chance to learn through practice (*bian zuo bian xue*). The cadres also organized on-the-ground technical lectures for the activists or recommended them for opportunities at provincial or county-level veterinary workshops. Local veterinary cadres in these ways helped some ordinary peasants become “the most reliable grassroots workforce” (*zuikekaode jiceng liliang*). These activists were the “wellspring of future [veterinary] cadres,” and as such one or two activists in each elementary cooperative or production team came to be officially titled epizootic prevention workers.⁴⁵

Let us take a closer look at how veterinary cadres and activists-turned-prevention workers worked together by examining a specific case from Beijing in 1952. When swine fever hit the capital in the early spring of that year, veterinary cadres mobilized 342 youth from forty-seven nearby villages and organized intensive six-day vaccination technical sessions. During this training period, they were able to inject swine fever vaccines in 3,745 pigs, 42.5 percent of the total 8,337 swine population in the participating villages. To vaccinate the remaining pigs, the class was split into work teams of four to twelve people and sent down to unvisited villages. While giving animals a shot, they put informative posters on billboards and explained the benefits of vaccination door to door (Figure 1).⁴⁶

⁴⁵ Ibid.

⁴⁶ “Zhuwen liuxing zhili qingkuang” [Situation on governing the spread of swine fever] (April 24, 1952), BMA, 009-002-00296-073.

在集体内部进行预防，兽医干部热衷于招募“积极分子” (*jijifenzi*)，即对共产党表现出强烈支持和忠诚的热情村民。他们鼓励这些积极分子参与牲畜的定期疫苗接种，并让他们有机会通过实践学习 (*bian zuo bian xue*)。干部们还为积极分子组织了实地技术讲座，或推荐他们参加省级或县级兽医研讨会。地方兽医干部通过这些方式帮助一些普通农民成为“最可靠的基层力量” (*zuikekaode jiceng liliang*)。这些积极分子是“未来 [兽医]干部”的“源泉”，因此每个初级合作社或生产队都有一两名积极分子被正式命名为疫病防治员。⁴⁵

让我们通过一个1952年北京的具体案例，更深入地了解兽医干部和由积极分子转变为防疫工作者的人员是如何协同工作的。当年初春，猪瘟袭击了首都，兽医干部动员了来自附近47个村庄的342名青年，组织了为期六天的强化疫苗接种技术培训。在培训期间，他们为3745头猪注射了猪瘟疫苗，占参与村庄8337头猪总数的42.5%。为了给剩余的猪接种疫苗，学员们被分成四到十二人的工作小组，并被派往尚未到访的村庄。在给动物注射疫苗的同时，他们在广告牌上张贴宣传海报，并挨家挨户解释疫苗接种益处（图1）。⁴⁶

⁴⁵ 同上。⁴⁶ “猪瘟流行治理情况” [猪瘟蔓延治理情况]（1952年4月24日），BMA, 009-002-00296-073。

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Figure 1. Certificates of swine vaccination from the 1970s. The main vaccinators were prevention workers, while the form of the certificate was designed and verified by the county veterinary station. Source: Author’s collection.

Some peasants, however, were still worried that the vaccine might harm their precious animals. Villagers on the outskirts of Beijing even openly demonstrated their mistrust in the technical competence of the teams, saying, “I will give you the vaccination fee if my pigs will be alive in six months.”⁴⁷ Guangxi peasants, too, were skeptical about vaccinators. They complained that after getting a vaccine, “most oxen got emaciated and lethargic, limped, and had diarrhea.” They therefore found vaccination to be disturbing agricultural work. Once a few of the vaccinated oxen came to be dead, they explicitly refused to get their livestock

⁴⁷ Ibid.

这是预印本，已接受发表，计划刊登于《医学史通报》第98卷第2期（2024年夏季刊）。本文已进行编辑校对，但尚未分页。可能还会进行进一步编辑。请留意最终文章的发布详情。



图1. 20世纪70年代的生猪疫苗接种证明。主要接种人员是防疫工作者，而证明的形式由县兽医站设计和验证。来源：作者收藏。

然而，一些农民仍然担心疫苗可能会伤害他们宝贵的牲畜。北京郊区的村民甚至公开表示不信任技术团队的能力，他们说：“如果我的猪能在六个月内活下来，我就付你疫苗接种费。”⁴⁷ 广西的农民也对疫苗接种人员持怀疑态度。他们抱怨说，接种疫苗后，“大多数牛都变得消瘦、嗜睡、跛行，并出现腹泻。”因此，他们认为疫苗接种扰乱了农业生产。一旦有几头接种过疫苗的牛死亡，他们就明确拒绝给他们的牲畜接种疫苗。

⁴⁷ 同上。

vaccinated.⁴⁸ Only after a great number of animals contracted diseases and the spread looked to be uncontrollable did the peasants call in veterinary workers and “direly asked for healing the infected animals.”⁴⁹ To make the CPT program fully work, the vaccinators—local veterinary cadres and prevention workers—had to gain their fellow peasants’ trust.⁵⁰

The cadres and prevention workers, however, lacked degrees from elite institutions or any other social capital to address this issue of trust. From the perspective of local peasants, the veterinary cadres, not to mention the peasants-turned-prevention workers, were by no means anonymous state agents arriving from above. Most were neighbors from the same or nearby villages. Furthermore, collectivization and the merging of several villages into a certain scale of collective meant these local actors shared a broader sense of community. Veterinary cadres and workers tried to form face-to-face ties with peasants and convince them to support vaccination.⁵¹

For instance, in December 1953, Pingle county in Guangxi, where 37,318 households,

⁴⁸ “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie” [Fuzhong county’s final work report on veterinary disease control in the spring of 1953] (April 30, 1953), GARA, X163-001-0256-0098.

⁴⁹ “Guanyu Wuzhou shi 1954 nian xumu shouyi gongzuo de baogao” [Report on the work of animal husbandry and veterinary medicine in the city of Wuzhou in 1954] (1954), GARA, X163-001-0395-0056.

⁵⁰ The problem of trust is inextricable from the construction of scientific knowledge beyond Maoist China. Simon Schaffer and Steven Shapin, *Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life* (1985; Princeton, N.J.: Princeton University Press, 2011); Theodore M. Porter, *Trust in Numbers: The Pursuit of Objectivity in Science and Public Life* (1995; Princeton, N.J.: Princeton University Press, 2020).

⁵¹ “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie” (n. 48). In her groundbreaking monograph on mass human vaccinations in early and mid-twentieth-century China, Mary Brazelton puts more stress on the state’s top-down, coercive measures than on grassroots trust networks. *Mass Vaccination* (n. 2), esp. chap. 6.

接种疫苗。⁴⁸ 只有在大量动物染病且疫情看起来无法控制之后，农民才请来兽医工作者，“迫切要求治疗受感染的动物。”⁴⁹ 为了使CPT项目充分发挥作用，疫苗接种者——当地兽医干部和防疫人员——必须赢得同乡农民的信任。⁵⁰

然而，这些干部和防疫人员缺乏来自精英机构的学位或任何其他社会资本来解决信任问题。在当地农民看来，兽医干部，更不用说由农民转变为防疫人员的那些人，绝不是从上级派来的匿名国家代理人。他们大多数是来自同村或附近村庄的邻居。此外，集体化和几个村庄合并成一定规模的集体意味着这些地方行动者拥有更广泛的社区意识。兽医干部和工作人员试图与农民建立面对面的联系，并说服他们支持疫苗接种。⁵¹

例如，1953年12月，广西平乐县有37318户家庭，

⁴⁸ “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie” [Fuzhong county’s final work report on veterinary disease control in the spring of 1953] (April 30, 1953), GARA, X163-001-0256-0098.

⁴⁹ “Guanyu Wuzhou shi 1954 nian xumu shouyi gongzuo de baogao” [Report on the work of animal husbandry and veterinary medicine in the city of Wuzhou in 1954] (1954), GARA, X163-001-0395-0056.

⁵⁰ The problem of trust is inextricable from the construction of scientific knowledge beyond Maoist China. Simon Schaffer and Steven Shapin, *Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life* (1985; Princeton, N.J.: Princeton University Press, 2011); Theodore M. Porter, *Trust in Numbers: The Pursuit of Objectivity in Science and Public Life* (1995; Princeton, N.J.: Princeton University Press, 2020).

⁵¹ “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie” (n. 48). In her groundbreaking monograph on mass human vaccinations in early and mid-twentieth-century China, Mary Brazelton puts more stress on the state’s top-down, coercive measures than on grassroots trust networks. *Mass Vaccination* (n. 2), esp. chap. 6.

39,723 cattle, and a larger number of pigs and chickens lived, was facing the outbreak of several animal diseases including anthrax, brucellosis, and swine erysipelas. Each collective within the county had its own group of veterinary cadres and activist-assistants to vaccinate farm animals. These veterinary workers “explained in detail the benefits of shots” as well as described anticipated immune reactions to the village elders whom they were familiar with for long. And about ten days after giving animals a shot, they made a “follow-up visit” (*fuchafangwen*) to collective farms. A peasant there told the cadres that “on the next day after the vaccination, some pigs did not eat and the spot where they got the shot swelled up. If you had not explained it to us beforehand, I would have been very worried.”⁵²

Similarly, veterinary workers in Fuzhong county, Guangxi, took advantage of their personal closeness to villagers who worked side by side with farm animals. From February 28 to March 31, 1953, a team of three veterinary cadres, a folk veterinarian, and an unknown number of prevention workers in charge of vaccinating livestock in their own cooperatives administered 583 rinderpest vaccines, 35 anthrax vaccines, 1,467 brucellosis vaccines, and 4,139 avian Newcastle disease vaccines. In the process, they encountered some “skeptics” and “bystanders” hanging around the vaccination sites. The team made sure to have them see for themselves that the vaccinated animals were unharmed and to elaborate the mechanism of vaccination and post-vaccinal issues.⁵³ To grassroots peasants, the trust in epizootic vaccines seemed to be little different from their trust in the vaccinators. Hence, the direct human relationship between peasants and veterinary workers within communities could increase the

⁵² “Pingle xian nonglin jishu zhidaozhan guanyu 1953nian xumu shouyi gongzuo zongjie” (n. 20).

⁵³ “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie” (n. 48).

39,723头牛以及更多的猪和鸡，正面临炭疽、布鲁氏菌病和猪丹毒等多种动物疾病的爆发。该县的每个集体都有自己的兽医干部和积极分子助理团队，负责给农场动物接种疫苗。这些兽医工作者“详细解释了注射的好处”，并向他们长期熟悉的村里长者描述了预期的免疫反应。在给动物注射大约十天后，他们会进行一次“复查访问”（*fuchafangwen*）集体农场。一位农民告诉干部，“接种疫苗的第二天，有些猪不吃东西，注射部位肿胀。如果你们没有事先解释，我一定会非常担心。”⁵²

同样，广西富钟县的兽医人员利用他们与农户的私人亲密关系，与农户并肩工作。1953年2月28日至3月31日，一个由三名兽医干部、一名民间兽医和数量不详的防疫人员组成的团队，负责为他们自己的合作社的牲畜接种疫苗，共接种了583剂牛瘟疫苗、35剂炭疽疫苗、1467剂布鲁氏菌病疫苗和4139剂禽新城疫疫苗。在此过程中，他们遇到了一些在接种点附近徘徊的“怀疑者”和“旁观者”。该团队确保让他们亲眼看到接种疫苗的动物安然无恙，并详细阐述了疫苗接种的机制和接种后的问题。⁵³ 对于基层农民来说，对动物疫病疫苗的信任似乎与他们对疫苗接种者的信任没有什么不同。因此，农民和社区内兽医人员之间直接的人际关系可以增加

⁵² “平乐县农林技术指导站关于1953年畜牧兽医工作总结”（n. 20）。⁵³ “富钟县1953年春季兽医防疫工作总结”（n. 48）。

general credibility of vaccination as well as the CPT program.

A decade later, in 1963, as local experiences accumulated continuously, the central Ministry of Agriculture reaffirmed the method of communal mass animal vaccination. Taking swine fever, for instance, the ministry claimed that more than ten years of preventative work verified the effectiveness of both vaccines and local vaccinators. China had already secured a valid vaccine technology, namely the rabbit-attenuated swine fever vaccine, which was easy and cheap to produce, quick to ensure immunity (four days after the vaccination), and long lasting (effective for a year and a half). Central leaders also stipulated that “each commune should train its full-time prevention workers, and each brigade needs to have a prevention worker.” And when epizootic and zoonotic diseases spread, each county would guide its communes and lower collectives to “combine the capabilities of veterinary cadres, veterinarians, and prevention workers” and organize a CPT taskforce. In doing so, grassroots communities could “take the three-in-one methodology incorporating local leaders, technicians, and the masses into the unfolding of the prevention and treatment work.”⁵⁴ Collective-led mass animal vaccination and its underlying principles of mass education were the foundation of Maoist China’s fight against contagious animal diseases.

⁵⁴ “Zhonghua renmin gongheguo nongyebu guanyu zai disange wunian jihua qinei jibenshang xiaomie zhuwen de zhishi” [PRC Ministry of Agriculture’s directive on basically eliminating swine fever during the Third Five-Year Plan period] (July 22, 1963), in *Xumu shouyi gongzuo cankao ziliao*, ed. Gansu sheng xumuting (Lanzhou: Gansu sheng xumuting, 1964), 51–52.

general credibility of vaccination as well as the CPT program.

十年后，即1963年，随着地方经验的不断积累，中央农业部重申了社队群众性畜禽疫苗接种的方法。以猪瘟为例，农业部声称，十多年的预防工作验证了疫苗和地方接种人员的有效性。中国已经掌握了有效的疫苗技术，即兔化弱毒猪瘟疫苗，这种疫苗生产简便、成本低廉、免疫迅速（接种后四天即可产生免疫力），且持续时间长（有效期一年半）。中央领导还规定，“每个公社都应培训其专职防疫人员，每个大队都需要有一名防疫人员。”当动物流行病和人畜共患病蔓延时，各县将指导其公社和下级集体“结合兽医干部、兽医和防疫人员的能力”，组织一个CPT工作组。通过这种方式，基层社区可以“将地方领导、技术人员和群众三结合的方法融入到防治工作的开展中。”⁵⁴集体主导的群众性动物疫苗接种及其群众教育的基本原则是毛泽东时代中国对抗动物传染病的基础。

⁵⁴ “中华人民共和国农业部关于在第三个五年计划期内基本上消灭猪瘟的指示” [中华人民共和国农业部关于在第三个五年计划期内基本上消灭猪瘟的指示]（1963年7月22日），载甘肃省畜牧厅编：《畜牧兽医工作参考资料》（兰州：甘肃省畜牧厅，1964年），第51-52页。

Producing Vaccines On Site

In the 1960s and 1970s, as the CPT program became the standard in response to animal diseases, the notion that each commune was primarily responsible for its livestock’s health was increasingly taken for granted. This even led to the idea that local collectives should be able to produce vaccines, especially swine fever vaccines, on their own without asking the upper-level state apparatus for help. The Ministry of Agriculture determined that when it came to the rabbit-attenuated swine fever vaccine, “local procurement” was the rule.⁵⁵ More specifically, it required each county to select one or two communes under its jurisdiction to establish “vaccine manufacturing posts” (*zhimiaodian*). Veterinary cadres on site would lead these institutions, which were expected to supply all the vaccines for the entire county. Each vaccine station should systematically raise enough fertile female rabbits to “guarantee the sources of the vaccine production.” The ministry also warned local leaders that having more than two vaccine manufacturing posts in their county should be discouraged. Although the rabbit-attenuated swine fever vaccine “is easy to produce, it is also easy to contaminate when it is not properly handled, which could have a huge impact.” Now local communes, following the planning of the county to which they were affiliated, were expected to be self-reliant even in the supply of swine fever vaccine.⁵⁶

The vaccine was indeed producible on site as long as local manufacturing posts could

就地生产疫苗

在20世纪60年代和70年代，随着CPT项目成为应对动物疾病的标准，每个公社主要负责其牲畜健康的观念越来越被认为是理所当然的。这甚至导致了地方集体应该能够自行生产疫苗，特别是猪瘟疫苗，而无需向更高层国家机构寻求帮助的想法。农业部规定，对于兔化猪瘟疫苗，“地方采购”是惯例。⁵⁵ 更具体地说，它要求每个县在其管辖范围内选择一两个公社建立“制苗点”（制苗点）。现场兽医干部将领导这些机构，这些机构预计将为整个县提供所有疫苗。每个疫苗站都应系统地饲养足够多的繁殖母兔，以“保证疫苗生产的来源”。该部还警告地方领导，不鼓励在其县内设立两个以上的制苗点。尽管兔化猪瘟疫苗“易于生产，但如果处理不当也易受污染，这可能会产生巨大影响”。现在，地方公社，遵循其所属县的规划，甚至在猪瘟疫苗的供应方面也应自力更生。⁵⁶

只要当地生产点具备条件，疫苗确实可以在当地生产

⁵⁵ Ibid., 53.

⁵⁶ “Zhonghua renmin gongheguo nongyebu guanyu xiaomie zhuwen jianmiezhhan de tongzhi” [PRC Ministry of Agriculture’s notification on the annihilation war against swine fever] (April 28, 1964), in *Shouyi gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan (N.p., 1975), 7–9.

⁵⁵ 同上，第53页。⁵⁶ “中华人民共和国农业部关于消灭猪瘟歼灭战的通知” [中华人民共和国农业部关于消灭猪瘟歼灭战的通知]（1964年4月28日），载陕西省畜牧兽医总站编：《兽医工作文件选编》（出版地不详，1975年），第7-9页。

acquire viral antigens and the needed number of rabbits. Grassroots veterinary cadres injected the classical swine fever virus into healthy rabbits. The virus would not kill rabbits under normal conditions and would cause only several types of thermal reactions. The “typical thermal reaction” refers to having twenty-four to forty-eight hours of a latent period before running a fever. Rabbits would have a temperature of one to two degrees Celsius higher than normal for more than eighteen hours. The “mild thermal reaction” shows twenty-four to seventy-two latent hours and more than twelve hours of a temperature higher by half a degree to one degree Celsius. The “questionable thermal reaction” has irregularly short or long latent and high temperature periods, which signal that the rabbit’s immune system may have responded to other pathogens than the injected swine fever virus. Only the rabbits that show the first two reactions can be used for the production of vaccine.⁵⁷

The appropriately infected rabbits can provide the swine fever vaccine in two ways. The easiest way was to extract blood from the heart. It was called “blood vaccine” (*xuemiao*) and could be directly injected into pigs to induce immunity. The other method that was trickier but more reliable was the “spleen-lymph vaccine” (*pilinmiao*). This type of vaccine was taken from the dissected rabbits’ spleen or mesenteric lymph nodes. The spleen-lymph vaccine had to be diluted one hundred times with saline solution or distilled water before use. Initially, the liver and kidney were used as well. But as these two organs turned out to be unstable in virus content, producing the vaccine from these organs was banned after 1963. In

⁵⁷ “Zhuwen tuhua ruodu yimiao zhizao ji jianyan guicheng” [Regulations on the production and verification of the rabbit-attenuated swine fever vaccine] (August 1963), in *Xumu shouyi gongzuo cankao ziliao*, ed. Gansu sheng xumuting (Lanzhou: Gansu sheng xumuting, 1964), 55–57.

获取病毒抗原和所需数量的兔子。基层兽医干部将经典猪瘟病毒注射到健康的兔子体内。在正常情况下，该病毒不会杀死兔子，只会引起几种热反应。“典型热反应”是指在发烧前有 24 到 48 小时的潜伏期。兔子的体温会比正常体温高 1 到 2 摄氏度，持续 18 小时以上。“轻度热反应”表现为 24 到 72 小时的潜伏期，体温比正常体温高 0.5 到 1 摄氏度，持续 12 小时以上。“可疑热反应”的潜伏期和高温期不规则地短或长，这表明兔子的免疫系统可能对注射的猪瘟病毒以外的其他病原体产生了反应。只有表现出前两种反应的兔子才能用于疫苗生产。⁵⁷

适当感染的兔子可以通过两种方式提供猪瘟疫苗。最简单的方法是从心脏中提取血液。它被称为“血苗” (*xuemiao*)，可以直接注射到猪体内以诱导免疫。另一种方法虽然更棘手但更可靠，那就是“脾淋苗” (*pilinmiao*)。这种疫苗取自解剖后的兔子的脾脏或肠系膜淋巴结。脾淋苗在使用前必须用生理盐水或蒸馏水稀释一百倍。最初，肝脏和肾脏也曾被使用。但由于这两种器官的病毒含量不稳定，1963年后禁止从这些器官生产疫苗。在

⁵⁷ 《猪瘟兔化弱毒疫苗制造及检验规程》[（1963年8月），载甘肃省畜牧厅编：《畜牧兽医工作参考资料》（兰州：甘肃省畜牧厅，1964年），第55-57页。

principle, these “on-site produced vaccines” (*jiudi zhimiao*) needed to be administered immediately. However, when necessary, they could be stored at zero to four degrees Celsius for seventy-two hours, at fifteen to twenty degrees for forty-eight hours, and at twenty to twenty-five degrees for twelve hours. Regardless of breed, size, age, sex, or pregnancy, the vaccine was administrable to all pigs. In terms of the injection site, the root of the ear or the buttocks were ideal for fully grown pigs and the inner thigh of the hind legs for piglets. Pigs would gain immunity four days after inoculation, and the immunity would last for a year and a half.⁵⁸

As it seemed feasible for the central and provincial governments to outsource to local collectives the production and administration of the rabbit-attenuated swine fever vaccine, they encouraged veterinary cadres and prevention workers to participate in extra training sessions to learn knowledge about the particular vaccine. For example, in 1964 prevention workers from all the communes and brigades in Gansu province were required to attend workshops on the making of the rabbit-attenuated swine fever vaccine organized by the prefecture- and county-level veterinary authorities. They were seven- to ten-day workshops and focused on the techniques for making and injecting the vaccine. The workshop attendees were then asked to go back to their collectives and find “the educated youths whose family background, thoughts, and labor attitude are good and who love veterinary work and have a certain level of cultural refinement,” thereby training more grassroots prevention workers and “making the technology take root in every production team.” Gansu provincial leaders could

⁵⁸ Ibid., 60–62.

原则上，这些“就地制苗” (*jiudi zhimiao*)需要立即使用。但在必要时，它们可以在0到4摄氏度下储存72小时，在15到20摄氏度下储存48小时，在20到25摄氏度下储存12小时。无论品种、大小、年龄、性别或是否怀孕，所有猪只都可以接种疫苗。就注射部位而言，成年猪的耳根或臀部是理想选择，仔猪则是后腿内侧。猪只在接种后四天即可获得免疫力，免疫力可持续一年半。⁵⁸

由于中央和省级政府将兔源猪瘟疫苗的生产和管理外包给地方集体似乎是可行的，因此他们鼓励兽医干部和防疫人员参加额外的培训课程，学习有关该疫苗的知识。例如，1964年，甘肃省所有公社和生产队的防疫人员都被要求参加由地市级和县级兽医部门组织的兔源猪瘟疫苗制作讲习班。这些讲习班为期七到十天，重点是疫苗制作和注射技术。讲习班的参与者随后被要求回到他们的集体，寻找“家庭背景、思想和劳动态度良好，热爱兽医工作并具有一定文化修养的知识青年”，从而培训更多的基层防疫人员，“使技术在每个生产队扎根”。甘肃省领导可以

⁵⁸ 同上，60-62。

remain optimistic in fighting swine fever since approximately twenty thousand prevention workers and more than five thousand veterinarians across the province were active in the “relatively sound grassroots veterinary organizations.”⁵⁹

Nevertheless, as Allan Brandt’s insight reminds us that “the promise of the magic bullet has never been fulfilled,”⁶⁰ swine fever was still rampant in the 1970s despite the grassroots production and accessibility of the vaccine. In neighboring Shaanxi province, for instance, collective-based mass vaccination continued to be emphasized in 1972 in tandem with the slogans of “self-breeding and self-raising” and “keeping sheds clean and disinfected.” The province proudly reported that 6,920,000 doses of the rabbit-attenuated swine fever vaccine were produced on site at the sub-county level and that twelve million pigs were vaccinated by grassroots veterinary workers within the year. Shaanxi recorded an 83 percent vaccination rate for the spring regular inoculation and 76.8 percent for the autumn. By the end of 1972, swine fever basically disappeared in 1,328 communes, which represented more than half of all the communes in the province. Despite these impressive numbers, which were known only to the provincial or upper bureaucrats, swine fever reemerged in May 1973. The disease was never eradicated during or after the Maoist era. Veterinary workers in communes and lower collectives kept doing their job.⁶¹

⁵⁹ “Guanyu jiben xiaomie zhuwen de yijian” (n. 26).

⁶⁰ Allan M. Brandt, *No Magic Bullet: A Social History of Venereal Disease in the United States Since 1880* (1985; Oxford: Oxford University Press, 1987), 161.

⁶¹ “Zhuanfa sheng xumu shouyi zongzhan guanyu jinkuai shixian quansheng jiben xiaomie zhuwen de yijian de tongzhi” [Notification on forwarding Shaanxi provincial veterinary station’s opinion on basically eliminating swine fever across the province as soon as possible] (May 4, 1973), in *Shouyi gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan (N.p., 1975), 14–18.

对防治猪瘟保持乐观，因为全省约两万名防疫人员和五千多名兽医活跃在“比较健全的基层兽医组织中”。⁵⁹

然而，正如艾伦·布兰特（Allan Brandt）的真知灼见提醒我们，“魔弹的承诺从未实现”，⁶⁰ 尽管疫苗在基层生产并易于获取，但猪瘟在20世纪70年代仍然猖獗。例如，在邻近的陕西省，1972年继续强调以集体为基础的大规模疫苗接种，同时提出“自繁自养”和“保持猪舍清洁消毒”的口号。该省自豪地报告称，在县级以下生产了692万剂兔源弱毒猪瘟疫苗，并在一年内由基层兽医人员为1200万头猪接种了疫苗。陕西省春季常规接种的疫苗接种率为83%，秋季为76.8%。到1972年底，猪瘟在1328个公社基本消失，这占全省公社总数的一半以上。尽管这些令人印象深刻的数字（仅为省级或上级官员所知），猪瘟在1973年5月再次出现。在毛泽东时代或之后，这种疾病从未被根除。公社和下级集体的兽医人员继续履行他们的职责。⁶¹

⁵⁹ “关于基本消灭猪瘟的意见”（n. 26）。⁶⁰ 艾伦·M·布兰特（Allan M. Brandt），《没有魔弹：1880年以来美国性病的社会史》（1985；牛津：牛津大学出版社，1987），161。⁶¹ “转发省畜牧兽医总站关于尽快实现全省基本消灭猪瘟的意见的通知” [Notification on forwarding Shaanxi provincial veterinary station’s opinion on basically eliminating swine fever across the province as soon as possible]（1973年5月4日），载陕西省畜牧兽医总站编《兽医工作文件选编》（无出版地，1975），14-18页。

Although this story of the on-site production of swine fever vaccine draws our attention to the proactive roles of local collectives in fighting animal infectious diseases, it is hard to determine the effectiveness of such vaccines. Above all, it is doubtful that the quality control was rigid. Maintaining and specifying groups of rabbits healthy enough for vaccine production must not have been easy or self-evident. Similarly, ensuring that every stage of the procedure for making vaccines from rabbits’ bodies was performed correctly would have been challenging. Also, given that the mass vaccination campaigns in the Maoist period for both humans and nonhumans tended to prioritize the sheer number of the vaccinated at any cost,⁶² and that local veterinary cadres often complained about the improper distribution of vaccines running out of shelf life,⁶³ it was highly likely that countless invalid vaccines were injected into communal livestock. Multispecies communal self-reliance was naturalized to the extent that one of the major epizootic vaccines was to be produced and used solely by local communes. But it came at a cost.

The Choice between Treatment and Culling

While we have discussed the preventative measures that represented the second character of the CPT, this section focuses on what the last character stood for: treatment. At different times and places than Maoist China, culling, instead of healing, might be the more common

⁶² Brazelton, *Mass Vaccination* (n. 2), esp. 123–43.

⁶³ “Pingle xian nonglin jishu zhidaozhan guanyu 1953nian xumu shouyi gongzuo zongjie” (n. 20); “Yulin zhuanqu 1970 nian jihua xuyao gezhong chuqin yijunmiao de baogao” [Yulin prefecture’s report on the necessity of various epizootic vaccines for its 1970 plans] (August 16, 1969), GYMA, 030-010-038-03.

尽管这个关于现场生产猪瘟疫苗的故事让我们关注到地方集体在对抗动物传染病方面的积极作用，但很难确定此类疫苗的有效性。最重要的是，其质量控制是否严格令人怀疑。维持和指定足够健康的兔群用于疫苗生产，想必并非易事或不言而喻。同样，确保从兔体制作疫苗的每个程序阶段都正确执行，也将是具有挑战性的。此外，考虑到毛泽东时代针对人类和非人类的大规模疫苗接种运动往往不惜一切代价优先考虑接种数量，⁶² 以及地方兽医干部经常抱怨疫苗分发不当导致过期，⁶³ 因此，无数无效疫苗很可能被注射到公社牲畜体内。多物种的公社自力更生被自然化到如此程度，以至于一种主要的动物疫病疫苗将完全由地方公社生产和使用。但这付出了代价。

治疗与扑杀的选择

我们已经讨论了代表CPT第二个字的预防措施，本节重点讨论最后一个字所代表的含义：治疗。在与毛泽东时代中国不同的时间地点，扑杀而非治疗可能更为常见

⁶² Brazelton, *Mass Vaccination* (n. 2)，特别是第123-43页。⁶³ “平乐县农林技术指导站关于1953年畜牧兽医工作总结”（n. 20）；“玉林专区1970年计划需要各种畜禽疫苗的报告” [玉林地区1970年计划所需各种动物流行病疫苗报告]（1969年8月16日），GYMA, 030-010-038-03。

option to take when historical actors fail to prevent the breakout of infectious animal diseases.⁶⁴ Especially in the West, the thorough culling of compromised animals has been regarded as the most efficient and economical way to control the diseases ever since Victorian England, although it has not gone entirely uncontested.⁶⁵

In contrast, local collectives in Maoist China were very conservative in slaughtering animals under all circumstances. On the ground, only veterinary cadres and veterinarians who worked at local veterinary stations could administratively decide which animals were to be killed. Without a slaughter permit issued by these local experts, in principle no one in the collectives could kill animals. Killable livestock were strictly limited to the “old, feeble, and disabled.”⁶⁶ This tendency hardly changed, even in the midst of the infamous Great Leap Famine in 1959–61. The Gansu Provincial Committee of the Communist Party in 1960 reaffirmed its commitment to “protecting people as well as livestock for the sake of production” (*baoren baochu, yili shengchan*) and required communes and lower collectives to “slaughter, systematically under the guidance of leaders, a certain number of sheep that

当历史参与者未能阻止传染性动物疾病爆发时可采取的选择。⁶⁴ 尤其是在西方，自维多利亚时代的英国以来，彻底扑杀受感染动物一直被认为是控制疾病最有效和最经济的方式，尽管这并非完全没有争议。⁶⁵

相比之下，毛泽东时代的中国地方集体在任何情况下都非常保守地对待宰杀动物。在基层，只有在地方兽医站工作的兽医干部和兽医才能行政决定哪些动物可以被宰杀。原则上，没有这些地方专家颁发的宰杀许可证，集体中的任何人都不能宰杀动物。可宰杀的牲畜严格限于“老、弱、病、残”。⁶⁶ 即使在 1959-61 年臭名昭著的大饥荒期间，这种趋势也几乎没有改变。1960 年，中共甘肃省委重申了“保人保畜，以利生产”（*baoren baochu, yili shengchan*）的承诺，并要求公社和下级集体“在领导的指导下，有计划地宰杀一定数量的羊，这些羊

⁶⁴ On the taboo of and sacred rituals regarding killing animals in premodern China as another exception, see Roel Sterckx, “Animal to Edible: The Ritualization of Animals in Early China,” in *Animals through Chinese History: Earliest Times to 1911*, ed. Roel Sterckx, Martina Siebert, and Dagmar Schäfer (Cambridge: Cambridge University Press, 2019), 46–63.

⁶⁵ On how this notion was historically constructed and has changed in and beyond England, see Abigail Woods, *A Manufactured Plague: The History of Foot and Mouth Disease in Britain* (London: Earthscan, 2004); John Law, “Culling, Catastrophe and Collectivity,” *Distinktion: Scandinavian J. Soc. Theory* 9, no. 1 (2008): 61–76.

⁶⁶ “Guangxi sheng nonglinting guanyu shouyi ren yuan ying jinkeneng zhixing zaisha niuzhi de jianyan qianzheng de hanfu” [Guangxi provincial department of agriculture and forestry’s response requiring veterinary workers to issue cattle slaughter inspection certificates as many as possible] (September 22, 1953), GARA, X163-001-0241-0015.

⁶⁴ 关于中国前现代时期杀戮动物的禁忌和神圣仪式，作为另一个例外，请参见罗尔·斯特克斯（Roel Sterckx），《动物到食物：早期中国动物的仪式化》，载于罗尔·斯特克斯、马丁娜·西伯特（Martina Siebert）和达格玛·舍费尔（Dagmar Schäfer）主编的《中国历史上的动物：从最早时期到1911年》（剑桥：剑桥大学出版社，2019年），第46-63页。⁶⁵ 关于这一概念在英国及其他地区是如何在历史上构建和变化的，请参见阿比盖尔·伍兹（Abigail Woods），《人造瘟疫：英国口蹄疫的历史》（伦敦：Earthscan，2004年）；约翰·劳（John Law），《扑杀、灾难与集体性》，载于《*Distinktion:Scandinavian J.Soc. Theory*》第9卷第1期（2008年）：第61-76页。⁶⁶ “广西省农林厅关于兽医人员应尽可能执行宰杀牛只的检疫签证的函复” [广西省农林厅关于兽医人员应尽可能执行宰杀牛只的检疫签证的函复]（1953年9月22日），GARA, X163-001-0241-0015。

need to be culled and give the mutton and lamb to the masses in famine areas, thereby increasing the meat supply and protecting the peoples’ physical strength.” Out of the 8,020,000 sheep and goats in the entire province, the committee allowed the cull of only 500,000 to 600,000, no more than 10 percent of the ovine population in pastoral districts and 7 percent in agricultural districts.⁶⁷

The hesitancy about killing collective animals was also found in the context of animal disease control. While promoting vaccination against swine fever, veterinary cadres ran into local peasants who regarded inoculated pigs’ loss of appetite as a sign of serious vaccine side effects. The villagers therefore asked them to issue a slaughter permit or even culled a few pigs themselves. The cadres had to “patiently educate them repetitively” that the vaccination should not be an excuse for the reckless slaughter of livestock.⁶⁸ Of course, there were times when culling appeared inevitable. However, for impoverished collectives, animals, if not animal bodies, were too precious to simply be discarded. Even though animal parts and tissue were compromised by epizootic diseases, they were still useful for the communal metabolic circle. Local veterinary workers and peasants cautiously sorted out the edible parts from culled or dead pigs that had suffered from relatively mild epizootics such as swine pneumonia or erysipelas. The meat blocks, liver, heart, lung, and kidney were deemed fit to eat if properly cooked. The stomach, intestine, bones, and blood could be selectively used as fertilizer. Only other useless parts were appropriately incinerated and buried

⁶⁷ “Shengwei guanyu zaisha yipi taotai yangzhi de tongzhi” [CCP Gansu Provincial Committee’s notification on culling a certain number of disposable sheep] (December 13, 1960), GPA, 192-005-1033-0016.

⁶⁸ “Pingle xian nonglin jishu zhidaozhan guanyu 1953nian xumu shouyi gongzuo zongjie” (n. 20).

需要扑杀，并将羊肉分发给饥荒地区的群众，从而增加肉类供应，保护人民的体力。”全省802万只绵羊和山羊中，委员会只允许扑杀50万至60万只，不超过牧区绵羊总数的10%，农业区绵羊总数的7%。⁶⁷

¹在动物疫病防治方面，也存在对宰杀集体动物的犹豫。在推广猪瘟疫苗接种时，兽医干部遇到当地农民，他们认为接种疫苗的猪食欲不振是疫苗严重副作用的迹象。因此，村民要求他们发放屠宰许可证，甚至自己宰杀了少数猪。干部不得不“反复耐心教育”他们，接种疫苗不应成为随意宰杀牲畜的借口。⁶⁸当然，有时扑杀似乎不可避免。然而，对于贫困的集体来说，动物，如果不是动物尸体，都太珍贵了，不能简单地丢弃。尽管动物的身体部位和组织受到流行病的损害，但它们对公共代谢循环仍然有用。当地兽医和农民小心翼翼地从不被扑杀或死亡的猪中分拣出可食用的部分，这些猪患有相对轻微的流行病，如猪肺炎或丹毒。肉块、肝脏、心脏、肺和肾脏如果烹饪得当，则被认为适合食用。胃、肠、骨骼和血液可以选择性地用作肥料。只有其他无用的部分才被适当焚烧和掩埋。

⁶⁷ “省委关于宰杀一批淘汰羊只的通知” [中共甘肃省委关于宰杀一批淘汰羊只的通知]（1960年12月13日），甘肃省档案馆，192-005-1033-0016。⁶⁸ “平乐县农林技术指导站关于1953年畜牧兽医工作总结”（注20）。

underground.⁶⁹

Moreover, there is evidence that the state policy toward mass culling of infected animals was twofold. On the one hand, central, provincial, and municipal governments were less reluctant to kill animals grown in urban distribution lines of animal protein, which typically comprised state-owned farms, temporary sheds run by municipal food companies, and supply and marketing cooperatives (*gongxiaoshe*) in major cities. But on the other hand, they were more prudential in culling pigs raised by and for local collectives. In September 1954, swine fever broke out, first in a farm owned by the Beijing municipal food company, and soon spread to other state-owned farms. By November, 4,200 pigs had died and 3,600 pigs were urgently culled when the city could not obtain vaccines in time. In December, the situation had not yet turned around, although more than 10,000 pigs had already been slaughtered. The Beijing Municipal Government eventually decided to cull the remaining 23,900 in the farms it managed directly.⁷⁰ In the same year, swine fever also swept through Guangxi, and the Guangxi Provincial Government similarly ordered the mass culling of pigs raised on state-owned farms for the purpose of providing meat to cities in the province and

⁶⁹ If pigs contracted anthrax, a more dangerous disease that can infect humans, the entire body had to be thoroughly burned out. “Dali fangzhi zhu chuanranbing de chubu yijian” (n. 20).

⁷⁰ “Zhuwen yanzhong” [The swine fever situation is serious] (November 16, 1954), BMA, 001-006-01027-00028; “Zhuwen jixu manyan” [Swine fever continues to prevail] (December 14, 1954), BMA, 001-006-01027-00029; “Fengtai qu fenfang siyang maozhu dabufen chuanran zhuyi siwang” [Most pigs raised in flour mills in Fengtai district have died of swine disease infection] (November 20, 1954), BMA, 046-002-00442-00238.

underground.⁶⁹

此外，有证据表明，国家对大规模扑杀受感染动物的政策是双重的。一方面，中央、省和市政府不太不愿意扑杀城市动物蛋白分销线中饲养的动物，这些分销线通常包括国有农场、市食品公司经营的临时棚屋以及主要城市的供销合作社（*gongxiaoshe*）。但另一方面，他们在扑杀由地方集体饲养并为其饲养的猪时则更为谨慎。1954年9月，猪瘟爆发，首先在北京食品公司下属的一个农场，很快蔓延到其他国有农场。到11月，4200头猪死亡，3600头猪在城市无法及时获得疫苗的情况下被紧急扑杀。到12月，尽管已有1万多头猪被宰杀，但情况尚未好转。北京市政府最终决定扑杀其直接管理的农场中剩余的23900头猪。⁷⁰ 同年，猪瘟也席卷广西，广西省政府也同样下令大规模扑杀国有农场饲养的猪，以向省内城市提供肉类，并且

⁶⁹ 如果猪感染了炭疽这种可能感染人类的更危险的疾病，则必须将整个猪体彻底焚烧。“大理防治猪传染病的初步意见”（n. 20）。⁷⁰ “猪瘟严重” [猪瘟疫情严重]（1954年11月16日），BMA, 001-006-01027-00028；“猪瘟继续蔓延” [猪瘟继续流行]（1954年12月14日），BMA, 001-006-01027-00029；“丰台区粉坊饲养毛猪大部分传染猪疫死亡” [丰台区面粉厂饲养的大部分猪死于猪瘟感染]（1954年11月20日），BMA, 046-002-00442-00238。

sending animal products to the nearby port of Guangzhou for export.⁷¹

However, the same document by the Guangxi government communicated different instructions to the local collectives. It required each collective to remain vigilant against the swine fever. Veterinary workers and the masses in the areas where the infection had not yet been detected needed to reinforce the general management of animal husbandry, especially keeping individual animals and their sheds clean and disinfected. In the areas already hit by the epizootic, in addition to promptly deploying quarantine and disinfection, the provincial government explicitly “prohibited [collectives] from indiscreetly killing infected pigs and circulating the untreated meat made of the infected.” These cases imply that collective-owned pigs were less killable than their state-owned counterparts. It was perhaps because the former were integral to the self-reliance and survival of grassroots collectives, whereas the latter were regarded as more easily disposable as the upper-level state apparatus was capable of replacing infected animals with healthy ones to maintain its stock for urban meat distribution.⁷²

If it was less likely that local veterinary workers culled pigs in their collectives, the same was the case for plowing oxen, which were usually more expensive and indispensable for agricultural production. A good example can be found in Gansu province, where FMD raged in 1964–65. As soon as they heard that the disease was spreading in neighboring Xinjiang, all people’s communes in the province embarked on an effort to vaccinate a total of

⁷¹ “Guangxi sheng nongyeting guanyu jiaqiang zhuwen fangzhi de tongzhi” [Guangxi provincial department of agriculture’s notification on strengthening the prevention and treatment against swine fever] (January 4, 1954), GARA, X163-001-0390-0077.

⁷² Ibid.

将动物产品运往附近的广州港出口。⁷¹

然而，广西政府的同一份文件向地方集体传达了不同的指示。它要求每个集体对猪瘟保持警惕。在尚未发现感染的地区，兽医工作者和群众需要加强畜牧业的总体管理，特别是保持个体动物及其棚舍的清洁和消毒。在疫情已经爆发的地区，除了及时部署隔离和消毒外，省政府明确“禁止 [集体] 随意宰杀受感染的猪，并流通未经处理的受感染猪肉。”这些案例表明，集体所有的猪比国有猪更不容易被宰杀。这或许是因为前者是基层集体自力更生和生存不可或缺的一部分，而后者则被认为更容易处理，因为上级国家机构能够用健康的动物替换受感染的动物，以维持其用于城市肉类分配的库存。⁷²

如果说当地兽医不太可能在其集体中扑杀生猪，那么耕牛也是如此，耕牛通常更昂贵，对农业生产来说不可或缺。甘肃省就是一个很好的例子，该省在1964-65年间爆发了口蹄疫。当地一听说疫情正在邻近的新疆蔓延，全省所有人民公社便开始着手为总计

⁷¹ “广西省农业厅关于加强猪瘟防治的通知” [广西省农业厅关于加强猪瘟防治的通知]（1954年1月4日），GARA, X163-001-0390-0077。⁷² 同上。

352,151 cattle. However, FMD eventually penetrated ninety-one communes and infected 17,737 oxen. Among them, 14,220 were bovine workers in collectives under Maqu county. Local cadres and veterinary workers in these collectives decided to isolate the infected from the healthy oxen and intensively heal and take care of the former rather than culling them out. That year, only eighty-four cattle died or were unavoidably slaughtered due to the epizootic, 0.5 percent of the number of the infected.⁷³ Given that the Maoist period is infamous for the unreliability of its statistical data, this information might not be fully trustworthy.⁷⁴ Still, it at least shows that grassroots collectives wanted to convince their superiors that they were successful in responding to FMD without losing too many of their cattle. I have not found any evidence that local veterinary workers were in fact enthusiastic about culling and covered it up in their reports to upper-level governments.

As we have seen, mass culling was not typically a part of the commune-based CPT program during the Maoist period. Treatment, not slaughtering, as its name indicates, was, along with prevention, at the core of the scheme. How then did local veterinary workers treat and cure livestock that had contracted epizootic and zoonotic diseases? The two main remedies were serums and traditional medicines. The Ministry of Agriculture compiled local communes' therapeutic practices and standardized the treatments for major animal infectious diseases in 1959. According to the handbook edited by the ministry and distributed to every commune, the FMD serum extracted from cattle recovered from the disease was a must-use.

⁷³ “Guanyu xiaomie shengchu koutiyi gongzuo zongjie baogao” (n. 28).

⁷⁴ On the development of statistics in the 1950s, see Arunabh Ghosh, *Making It Count: Statistics and Statecraft in the Early People's Republic of China* (Princeton, N.J.: Princeton University Press, 2020).

352,151 头牛。然而，口蹄疫最终蔓延到 91 个公社，感染了 17,737 头牛。其中，14,220 头是玛曲县集体所有的役牛。这些集体的当地干部和兽医决定将受感染的牛与健康的牛隔离，并对前者进行集中治疗和护理，而不是将其扑杀。那一年，只有 84 头牛因疫情死亡或被迫宰杀，占受感染牛数量的 0.5%。⁷³ 鉴于毛泽东时代统计数据不可靠而臭名昭著，这些信息可能并非完全可信。⁷⁴ 尽管如此，它至少表明基层集体希望让上级相信他们在应对口蹄疫方面取得了成功，而没有损失太多牛。我没有发现任何证据表明当地兽医实际上热衷于扑杀，并在向上级政府的报告中掩盖了这一点。

正如我们所见，在毛泽东时代，大规模扑杀通常不是公社兽医防疫计划的一部分。顾名思义，治疗而非屠宰，与预防一起，是该计划的核心。那么，地方兽医工作者如何治疗和治愈感染了动物流行病和人畜共患病的牲畜呢？两种主要的治疗方法是血清和传统药物。农业部于1959年汇编了地方公社的治疗实践，并对主要动物传染病的治疗方法进行了标准化。根据农业部编辑并分发给每个公社的手册，从康复的牛身上提取的口蹄疫血清是必须使用的。

⁷³ “关于消灭牲畜口蹄疫工作总结报告”（注28）。⁷⁴ 关于20世纪50年代统计学的发展，参见 Arunabh Ghosh, 《*Making It Count: Statistics and Statecraft in the Early People's Republic of China*》（普林斯顿，新泽西州：普林斯顿大学出版社，2020年）。

When it came to anthrax, two subcutaneous or intravenous injections of 150 to 300 milliliters of the serum with an interval of ten hours was the most common method to heal herds. It was complemented by 1,000 to 2,000 units of penicillin per kilogram when available. Likewise, swine fever serum was widely used for treating the disease. The recommended dose was 100 milliliters of the serum per 50 kilograms of pigs, with the second shot to follow the next day. Other than penicillin and sulfanilamide, all other drugs had proved ineffective.⁷⁵

Serums were effective, yet expensive. As of 1950, 1 milliliter of the swine fever serum was equivalent to 500 grams of millet.⁷⁶ Therefore, not surprisingly, central and provincial governments asked local communities to find a way to take advantage of what was available on the ground, which, in most cases, meant “using traditional medicines and trying experimental treatments.”⁷⁷ In 1959, for example, Tantang Commune of Guixian county, Guangxi, just like other grassroots communities, launched a mass scientific experiment campaign.⁷⁸ As a part of the campaign, thousands of youths, peasants, and veterinary workers worked to familiarize themselves with livestock artificial insemination technologies and experimented with traditional Chinese medicines they hoped would be effective against

对于炭疽病，最常见的治疗方法是皮下或静脉注射150至300毫升血清，间隔十小时注射一次。如果条件允许，每公斤体重辅以1,000至2,000单位青霉素。同样，猪瘟血清也广泛用于治疗该病。推荐剂量为每50公斤猪注射100毫升血清，第二天再注射第二针。除了青霉素和磺胺类药物，所有其他药物都被证明无效。⁷⁵

血清有效，但价格昂贵。截至1950年，1毫升猪瘟血清相当于500克小米。⁷⁶ 因此，中央和省级政府要求地方社区利用当地现有资源，这在大多数情况下意味着“使用传统药物并尝试实验性治疗”，这并不令人意外。⁷⁷ 例如，1959年，广西贵县的坦塘公社，像其他基层社区一样，发起了一场群众性科学实验运动。⁷⁸ 作为运动的一部分，成千上万的青年、农民和兽医工作者努力熟悉牲畜人工授精技术，并试验他们希望对疾病有效的传统中药，以对抗

⁷⁵ Nongyebu xumu shouyiju, *Renmin gongshe shouyi gongzuo shouce* (n. 17), 48, 52, and 69.
⁷⁶ “Beijing shi jiaoqu gongzuowei bagao jiaoqu zhuwen fasheng qingkuang” [Beijing suburban work committee’s report on the outbreak of swine fever in suburban areas] (March 24, 1950), BMA, 009-001-00125-00007.
⁷⁷ “Jieshao 51 nian koutiyi fangzhi jingyan” (n. 29).
⁷⁸ For more on the mass scientific experiment movement, see Schmalzer, *Red Revolution, Green Revolution* (n. 4), 3–6.

⁷⁵ 农业部畜牧兽医局，《人民公社兽医工作手册》（第17期），第48、52、69页。⁷⁶ “北京市郊区工作委报告郊区猪瘟发生情况” [北京市郊区工作委员会关于郊区猪瘟疫情爆发的报告]（1950年3月24日），北京市档案馆，009-001-00125-00007。⁷⁷ “介绍51年口蹄疫防治经验”（第29期）。⁷⁸ 有关群众性科学实验运动的更多信息，请参见Schmalzer，《红色革命，绿色革命》（第4期），第3-6页。

swine fever.⁷⁹

The case of the commune in Guangxi was hardly unique. More provinces and municipalities, including Beijing, Shanghai, Hebei, Shanxi, Shandong, Sichuan, Zhejiang, Fujian, and Yunnan pushed their communes and lower collectives to test traditional remedies and collect local medicinal herbs to treat pigs.⁸⁰ As was the case with the on-site rabbit-attenuated swine fever vaccine, the demand to secure effective traditional veterinary medicines locally was justified in the name of communal self-reliance. Some additional slogans such as “self-collecting, self-manufacturing” (*zicai zizhi*) and “self-producing, self-using” (*zichan ziyong*) appeared in official documents.⁸¹ In particular, Sichuan’s Wusheng county was touted by China’s biggest state media, *People’s Daily*, as exemplary in this regard. In 1959 folk veterinarians, veterinary cadres, prevention workers, and the masses from its thirty-five different communes eventually “discovered” about ninety “wild medicinal materials” that proved effective for treating pig diseases. The state-sponsored newspaper

⁷⁹ Chong Guo, “Tansuo zengchan de tujing: Tantang gongshe kexue yanjiu xingcheng qunzhong yundong” [Exploring ways to increase production: Scientific research in Tantang communes developed into a mass movement], *Renmin Ribao*, December 4, 1959.

⁸⁰ “Zhenjiu zhiliao zhuwen chenggong” [Successful treatment of swine fever with acupuncture and moxibustion] (February 14, 1960), BMA, 001-014-00609-00003; “Zhuanfa zhongyao zhiliao zhu ji bingzheng de jingyan” [Forwarding the experience of Chinese herbal medicine in treating pig and chicken diseases] (May 6, 1960), BMA, 119-001-00486-00012; “Jiangxi sheng nongyeting zhongshouyi shiyansuo 1957 nian shiyan zhongyao peifang jianjie” [The traditional Chinese veterinary medicine laboratory of Jiangxi provincial department of agriculture briefly introduces the experiments with Chinese medicinal formulas], *Minjian shouyi tongxun* 4 (March 1958): 20–21; “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushu” (n. 25).

⁸¹ “Wusheng xian jiben xiaomie zhude zhuyao yibing” [Wusheng county has basically eliminated major infectious swine diseases], *Renmin Ribao*, February 2, 1960; “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushu” (n. 25).

猪瘟。⁷⁹

广西公社的案例绝非个例。包括北京、上海、河北、山西、山东、四川、浙江、福建和云南在内的更多省市，都推动其公社和下级集体测试传统疗法并收集当地药草来治疗猪。⁸⁰ 与现场兔源弱毒猪瘟疫苗的情况一样，以公社自力更生的名义，对当地有效传统兽药的需求是合理的。一些额外的口号，如“自采自制” (*zicai zizhi*) 和“自产自用” (*zichan ziyong*) 出现在官方文件中。⁸¹ 特别是，四川武胜县被中国最大的官方媒体《人民日报》誉为这方面的典范。1959 年，来自其三十五个不同公社的民间兽医、兽医干部、防疫工作者和群众最终“发现”了大约九十种对治疗猪病有效的“野生药材”。国家资助的报纸

⁷⁹ Chong Guo, “Tansuo zengchan de tujing: Tantang gongshe kexue yanjiu xingcheng qunzhong yundong” [探索增产的途径：坦塘公社科学研究形成群众运动], 《人民日报》，1959年12月4日。⁸⁰ “Zhenjiu zhiliao zhuwen chenggong” [针灸治疗猪瘟成功] (1960年2月14日)，BMA, 001-014-00609-00003; “Zhuanfa zhongyao zhiliao zhu ji bingzheng de jingyan” [转发中草药治疗猪鸡疾病的经验] (1960年5月6日)，BMA, 119-001-00486-00012; “Jiangxi sheng nongyeting zhongshouyi shiyansuo 1957 nian shiyan zhongyao peifang jianjie” [江西省农业厅中兽医实验室1957年试验中药配方简介], 《民间兽医通讯》4 (1958年3月)：20-21; “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushu” (注25)。⁸¹ “Wusheng xian jiben xiaomie zhude zhuyao yibing” [武胜县基本消灭猪的主要疫病], 《人民日报》，1960年2月2日; “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushu” (注25)。

article proudly noted that thanks to such communal bottom-up efforts, not a single pig in the county died from swine fever that year, further consolidating the ideal of multispecies communal self-reliance.⁸²

Conclusion

Local veterinary workers—namely, lower-ranking veterinary cadres, folk veterinarians, epidemic prevention workers, and caretakers in communes, brigades, and teams— spearheaded the Comprehensive Prevention and Treatment against contagious animal diseases. Sticking to the priorities that maximized the extraction of resources from and minimized the material investment in the rural area, upper-level governments outsourced to the communal actors the tasks of carrying out lockdown, quarantine, and disinfection, vaccinating countless animals, and healing rather than culling infected animals. To protect communally shared livestock, which were integral in sustaining the life of the communes, local collectives had no choice but to find, make, and learn whatever was available and useful on the ground. And the goal of their fight against infectious animal diseases was not reduced to control pathogens in the most efficient and rapid way. They sought to maintain and sustain their communal capacity and integrity, which included nonhuman animals. In the course of deploying the CPT for about three decades, multispecies communal self-reliance was gradually naturalized in Maoist China.

After Mao’s death in 1976, the social experiment of collective economy ended, and

文章自豪地指出，多亏了这种自下而上的集体努力，当年该县没有一头猪死于猪瘟，这进一步巩固了多物种集体自力更生的理想。⁸²

结论

地方兽医工作者——即基层兽医干部、民间兽医、防疫人员以及公社、大队和生产队的饲养员——率先开展了动物传染病综合防治工作。为了最大限度地从农村地区提取资源，并最大限度地减少对农村地区的物质投入，上级政府将封锁、隔离、消毒、为无数动物接种疫苗以及治疗而非扑杀受感染动物的任务外包给公社行动者。为了保护对维持公社生活至关重要的集体牲畜，地方集体别无选择，只能在当地寻找、制造和学习一切可用和有用的东西。他们与动物传染病作斗争的目标并不仅仅是为了以最有效和最快速的方式控制病原体。他们寻求维持和保持其集体能力和完整性，其中包括非人类动物。在大约三十年的综合防治部署过程中，多物种集体自力更生 in 毛泽东时代的中国逐渐被自然化。

1976年毛泽东逝世后，集体经济的社会实验宣告结束，

⁸² “Wusheng xian jiben xiaomie zhude zhuyao yibing” (n. 81).
38

⁸² “武胜县基本消灭猪的猪瘟”（注81）。
38

people’s communes and lower collectives across the nation were gradually dismantled between 1978 and 1985. The CPT, too, came to an end. Some former barefoot veterinarians and prevention workers took advantage of their expertise and became entrepreneurial animal doctors in their villages. There were no longer collective animal farms or communally owned livestock. Individual animal owners paid for veterinary service. Accordingly, the sense of communal self-reliance and common interests started to slip away. If an epizootic disease spread in a region, individuals would take responsibility only for their own stock. Reflecting such postsocialist changes, the Guangxi Zhuang Autonomous Region Government promulgated new legislation on the health of livestock and poultry in 1980. It specified the transfer of responsibility for epizootic and zoonotic disease control from collectives to individual livestock owners. For the first time, as far as I know, culling and slaughtering came to be overtly included in juxtaposition with “treating infected animals in quarantine” in this kind of official document in Guangxi.⁸³

This story of the CPT could be seen as another tragicomedy of communist absurdity. The ideal of multispecies communal self-reliance did not always work well, nor was it always ideal. It tells us as much about the irresponsibility of the Chinese Communist state as the agency of communal actors. The state, after all, failed to come up with the ways in which resources were distributed for multispecies communal well-being on a national scale. Nevertheless, rather than ridiculing the experience of Maoist China, we can criticize it from a

⁸³ “Guangxi Zhuangzu zizhi qu jiachu jiaqin jianyi zanxing guize” [Guangxi Zhuang Autonomous Region’s preliminary rules on livestock and poultry disease control] (March 29, 1980), in *Guangxi shouyi fangjian tongxun*, ed. Guangxi shouyi fangjianzhan (Nanning: Guangxi qumaoqiao yinshuachang, 1981), 1–3.

1978年至1985年间，全国各地的人民公社和低级集体逐渐解体。CPT也随之终结。一些前赤脚兽医和防疫工作者利用他们的专业知识，成为村里的创业型动物医生。不再有集体动物农场或公有牲畜。个体动物主人支付兽医服务费用。因此，集体自力更生和共同利益的意识开始消退。如果流行病在一个地区蔓延，个人只对自己的牲畜负责。为了反映这种后社会主义的变化，广西壮族自治区政府于1980年颁布了新的畜禽卫生立法。它明确规定将流行病和人畜共患病控制的责任从集体转移到个体牲畜主人。据我所知，这是广西此类官方文件中首次明确将扑杀和屠宰与“隔离治疗受感染动物”并列提及。⁸³

CPT 的故事可以被视为又一出共产主义荒诞剧。多物种群落自力更生的理想并非总是奏效，也并非总是理想的。它既揭示了中国共产主义国家的不负责任，也揭示了群落行动者的能动性。毕竟，国家未能提出在全国范围内分配资源以实现多物种群落福祉的方法。

然而，与其嘲笑毛泽东时代的中国经验，不如从一个

⁸³ “广西壮族自治区家畜家禽检疫暂行规定” [广西壮族自治区家畜家禽疾病控制暂行规定]（1980年3月29日），载《广西兽医防疫通讯》，广西兽医防疫站编（南宁：广西区毛桥印刷厂，1981），第1-3页。

perspective that increasingly values the idea that humans and nonhumans coinhabit a shared ecology, if not a commune, of the planet. The ideal of multispecies communal self-reliance in Mao-era China made individual rural residents feel in their bones that their subsistence was inevitably connected to the health of animals. They therefore tried their best to keep their invaluable animal companions alive and healthy with their knowledge and labor under given conditions so that they could preserve their own multispecies communities. If so, would it not be a stretch to say that envisioning a more sustainable earth, we can at least learn such a sense of communal responsibility from Chinese veterinary workers?

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JONGSIK CHRISTIAN YI is an assistant professor of the history of science in the Division of the Humanities and Social Sciences at Pohang University of Science and Technology (POSTECH), South Korea. He is currently working on his first monograph, *More-Than People’s Communes: Veterinary Workers and Nonhuman Animals in Mao-Era China*, which is under contract with the University of Chicago Press.

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这种观点越来越重视人类和非人类共同居住在一个共享生态（如果不是一个公社）的星球上的理念。毛泽东时代中国多物种公社自力更生的理想让个体农村居民深切感受到他们的生计与动物的健康密不可分。因此，他们尽最大努力在既定条件下，用他们的知识和劳动来保持他们宝贵的动物伙伴的生命和健康，以便他们能够维护自己的多物种社区。如果是这样，那么说在构想一个更可持续的地球时，我们至少可以从中国兽医工作者那里学习这种集体责任感，难道不是一种夸大吗？

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JONGSIK CHRISTIAN YI 是韩国浦项科技大学 (POSTECH) 人文社会科学部科学史助理教授。他目前正在撰写他的第一部专著《不止于人民公社：毛泽东时代的兽医工作者与非人类动物》，该书已与芝加哥大学出版社签约。

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